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The multi-commodity vehicle routing problem with temperature requirements

Rosario Paradiso [a,*], Said Dabia [a], Wout Dullaert [a], Joaquim Gromicho [b]

[a] Department of Operations Analytics, Vrije Universiteit Amsterdam, De Boelelaan, Amsterdam, 1081 HV, The Netherlands
[b] Faculty of Economics and Business, University of Amsterdam, Roetersstraat, Amsterdam, 1018 WB, The Netherlands

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ABSTRACT

Refrigerated transport requires distributing products with heterogeneous temperature constraints using vehicles equipped with cooling systems. Beyond traditional routing and capacity decisions, logistics providers must also determine the target temperature inside each vehicle. In this setting, each commodity must be transported within a prescribed temperature interval, so the chosen target temperature directly affects feasibility and determines which commodities can be jointly loaded on a vehicle. We introduce the Multi-Commodity Vehicle Routing Problem with Temperature Requirements (MC-VRP-T) which integrates temperature decisions into the routing process. We study two variants of this problem: (i) a constant-temperature variant, in which each vehicle selects a single temperature that is maintained throughout the route. For this variant, we develop a classical vehicle-indexed formulation and a novel temperature-indexed formulation exploiting an identified structural property, both solved via branch-and-cut, and we further embed this property in a set-partitioning formulation solved with a branch-and-cut-and-price algorithm. (ii) a temperature-adjustable variant, in which a vehicle may adjust its target temperature after each stop. For this more flexible setting, we provide a dedicated mathematical formulation solved via branch-and-cut and compare its complexity and benefits with the constant-temperature variant. The objective in both variants is to jointly determine routes and temperature settings that satisfy commodity-specific temperature requirements while minimizing routing and temperature-related costs. Computational experiments evaluate the performance of the formulations and quantify the economic impact of integrating routing and temperature decisions in refrigerated distribution.

1. Introduction

Transportation and distribution activities play a crucial role in modern supply chains. Efficient transportation and distribution planning can increase customer satisfaction and profitability and reduce costs. At the same time, transportation has important environmental consequences, causing 14% of the total worldwide CO_2 emissions (Dekker et al., 2012). The situation is even more severe for transporting perishable products (e.g., fresh and frozen food, pharmaceuticals). This stems from the fact that additional energy is needed to regulate temperature to ensure quality and limit waste during distribution. The additional thermal energy required is significant since it ranges from 15% to 25% of the motive energy of trucks (Tassou et al., 2009).

* Corresponding author.
E-mail addresses: r.paradiso@vu.nl (R. Paradiso), s.dabia@vu.nl (S. Dabia), w.e.h.dullaert@vu.nl (W. Dullaert), j.a.s.gromicho@uva.nl (J. Gromicho).

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Temperature-sensitive products typically have to be transported under different temperature regimes. Trucks are equipped with cooling/heating units and sensors to ensure the desired temperature for the cargo. The desired temperature inside the cargo space, referred to as “target temperature", must be set in accordance with the temperature requirements of the consolidated products (e.g., mushrooms can be transported at temperatures between 1 - 4 ° C, while eggplants between 5 - 9 ° C (Aung and Chang, 2014)).

Although Vehicle Routing has been an active field of research for five decades (Toth and Vigo, 2014), the optimization of temperature-controlled transportation has more recently received academic attention due to its environmental and economic impact (e.g., Mishra et al., 2016; Stellingwerf et al., 2018, 2021; Deng et al., 2022). The works addressing Vehicle Routing Problems (VRPs) for temperature-controlled products do not consider decisions on the cargo target temperature. Their use may, therefore, result in sub-optimal or infeasible decisions for products with different temperature requirements. Moreover, the vast majority of the literature considers a single commodity, whereas customers may request products with different temperature requirements as indicated by Aung and Chang (2014).

Inspired by a real-life case in refrigerated transport, we examine the benefits of integrating routing and temperature control decisions in a multi-commodity setting and study the different trade-offs involved. In this setting, each commodity must be transported within a prescribed temperature interval, so the chosen target temperature directly affects feasibility and determines which commodities can be jointly loaded on a vehicle. We are therefore the first to propose the Multi-Commodity Vehicle Routing Problem with Temperature Requirements (MC-VRP-T) that minimizes costs due to transportation and refrigeration by deciding on target temperatures and routing decisions in an integrated manner. For this problem, we propose a classical vehicle-index formulation and a new temperature-index formulation exploiting a novel structural property, both solvable via branch-and-cut. We leverage the same structural property in a set partitioning model solvable with a branch-and-cut-and-price algorithm. Our new temperature flow formulation and the branch-and-price-and-cut algorithm are capable of solving instances up to 100 customers. Moreover, we examine the impact of adjusting target temperatures along the routes by introducing and studying an extension of the MC-VRP-T, the MC-VRP-T with Adjustable Temperature (MC-VRP-TA), for which we develop a dedicated mathematical formulation and investigate how temperature adjustability influences both the total cost and the structure of the resulting distribution plans.

The remainder of the paper is structured as follows: Section 2 presents a focused literature review, and Section 3 introduces the MC-VRP-T and a vehicle flow formulation. Section 4 studies the structure of the proposed MC-VRP-T and presents alternative formulations. Section 5 is dedicated to the MC-VRP-TA. Extensive computational results are reported in Section 6. Conclusions and directions for further research are provided in Section 7.

2. Literature review

In Section 2.1, we review vehicle routing problems with perishable products, distinguishing between single-commodity and multi-commodity settings, and summarizing how quality decay and temperature control are modeled in the literature. Section 2.2 discusses incompatibility constraints in vehicle routing, focusing on cases where product characteristics-such as temperature requirements or hazard risks-prevent co-transportation, and outlines solution approaches for these constraints.

2.1. Vehicle routing problems with perishable products

Driven by growing environmental and economic concerns, the transportation of perishable goods within cold chain logistics has garnered significant attention in the vehicle routing literature. In the case of a **single perishable commodity**, Hsu et al. (2007) address a VRP with Time Windows (VRPTW) with stochastic travel times that influence product spoilage and present a nearest neighbor heuristic to solve it. Osvald and Stirn (2008) consider time-dependent travel times and quality loss, assuming linear deterioration of product quality over time, and solve the problem via a tabu search-based heuristic. Rong et al. (2011) integrate production and distribution planning within a food supply chain context, incorporating linear and exponential quality decay functions that depend on time and temperature. These decay functions are embedded in a mixed-integer programming model and solved using a general-purpose solver. Qi and Hu (2020) propose a hybrid heuristic approach combining an Ant Colony and Pareto Local Search to minimize fuel, refrigeration, and time-dependent quality degradation costs. Stellingwerf et al. (2018, 2021) employ a kinetic model that accounts for time- and temperature-dependent quality decay in fresh food distribution, highlighting trade-offs among routing cost, emissions, network structure, and product quality. Their findings suggest that minimizing cost or emissions can result in longer routes and increased quality degradation. Yu et al. (2022) address a combined order selection and periodic VRPTW for perishable products in the context of online community group-buying. They develop a mathematical model and propose a branch-and-price algorithm, validated through a real-world case study from a Chinese operator. Babaee et al. (2022) study an integrated pricing and transportation problem in a perishable product supply chain. They propose a nonlinear optimization model and solve it using a Benders-type decomposition approach that circumvents the need for solving subproblems directly. Their results reveal a non-monotonic relationship between transportation cost and product waste, and indicate that lower customer sensitivity to quality leads to higher waste-to-sales ratios. Ding et al. (2025) propose a target-oriented robust optimization model for cold chain routing problems, aiming to preserve product freshness and reduce carbon emissions under uncertainty simultaneously. Their approach incorporates both freshness targets and carbon budgets directly into the model constraints, illustrating the advantages of jointly addressing environmental and product quality objectives within a unified optimization framework.

In the case of **multiple perishable commodities**, Chen et al. (2009) consider the integration of production and routing decisions under stochastic demand and heterogeneous quality decay rates across products. The problem is modeled as a nonlinear program, with a constrained Nelder-Mead method solving the production layer and a heuristic algorithm addressing the routing layer. Amorim

and Almada-Lobo (2014) propose a multi-objective VRP framework that balances routing efficiency with freshness, assuming linear deterioration of product quality. They employ a hybrid evolutionary algorithm for solution purposes. Song and Ko (2016) consider heterogeneous vehicle types (refrigerated and non-refrigerated) in a food delivery context, modeling customer satisfaction as a function of freshness at delivery. More recently, Deng et al. (2022) examine a pickup-and-delivery VRPTW for perishable products, incorporating quality decay and refrigeration costs. They develop a branch-and-price algorithm and additionally optimize departure times from the depot. Tsang et al. (2024) formulate a joint optimization of sustainable order packing and multi-temperature delivery problem, and developed a metaheuristic solution approach to demonstrate both cost savings and higher customer satisfaction. Lin et al. (2025) introduce a dynamic VRP model for precooling logistics that integrates temperature-dependent quality decay into the routing decision. They account for the fact that perishable product quality deteriorates differently depending on time, temperature exposure, and precooling dynamics, and they model this using dynamic quality decay functions. Their results emphasize the importance of synchronizing routing and precooling operations to maintain product freshness and minimize waste. For a recent review of routing problems for perishable products, we refer the reader to Guerriero et al. (2025).

2.2. Incompatibility in vehicle routing problems

In practice, customer demands often span multiple commodities, some of which are mutually incompatible and thus cannot be co-transported. Examples include the separation of food and chemical goods or distinct categories of waste. Reviews of multi-commodity VRPs are provided in Archetti et al. (2015, 2016), Ferreira et al. (2025).

Product incompatibility imposes additional constraints on transportation planning and often leads to higher costs due to the increased number of required vehicles. Several studies have addressed the issue of incompatibility in transportation problems, particularly within the context of vehicle routing problems (VRPs). For instance, Gendreau et al. (2016) examine a setting where multiple pairs of products with unitary demand are mutually incompatible and, as such, cannot be transported together in the same vehicle. They employ a column generation-based solution method. Manerba and Mansini (2015) generalize this problem to allow for larger order quantities and solve it using a branch-and-cut algorithm. Ceselli et al. (2009) propose a rich VRP formulation that includes incompatibility constraints between products, depots, vehicles, and customers, addressed through a column generation-based algorithm. Battarra et al. (2009) consider a multi-trip VRP with incompatibility constraints, minimizing the number of vehicles used through an iterative algorithm. Factorovich et al. (2020) analyze a single-vehicle pickup and delivery problem under incompatibility constraints and develop three integer programming formulations, selecting the most effective for a branch-and-cut solution. Wu and Li (2024) investigate a multi-compartment truck-and-trailer VRP for petrol station replenishment involving domino hazard risks. They propose a decomposition-based matheuristic with improved upper and lower bounds and account for petrol evaporation during transportation, inventory storage, and inter-compartment transfer. Their work highlights the importance of modeling physical phenomena specific to hazardous and perishable goods in routing decisions.

In the context of perishable goods, incompatibility often arises from divergent temperature requirements. Deng et al. (2022), for instance, incorporate incompatibility into a pickup-and-delivery routing problem and treat vehicle temperature as a decision variable. Luo et al. (2025) study a time-dependent pickup and delivery problem within cold chain logistics that includes product incompatibility constraints. They propose an exact branch-price-and-cut algorithm capable of handling time-dependent travel durations and incompatibility requirements, and show the effectiveness of their method through computational experiments.

Incompatible products can be transported using the same vehicle if it is or can be compartmented. Such multi-compartment vehicles (MCVs) have several applications, including, e.g., the distribution of petroleum products to petrol stations and the collection of various waste types. In the literature, several VRPs involving MVCs are solved. This family of problems is known as the Multi-Compartment Vehicle Routing Problem (MCVRP) and was first introduced by Brown and Graves (1981). Several variants of the MCVRP are introduced depending on the characteristics of the vehicles' compartments. There are problems with fixed compartment size, as in, e.g., Hsu et al. (2007), Coelho and Laporte (2015), Wu and Li (2024), versus problems with flexible compartment size as in, e.g., Henke et al. (2015), Henke et al. (2019). Moreover, there are problems with fixed assignment of product types to compartments (e.g., van der Bruggen et al., 1995; Coelho and Laporte, 2015; Yahyaoui et al., 2020) versus problems with flexible assignment of product types to compartments (e.g., Derigs et al., 2011; Archetti et al., 2015; Wang and Li, 2018). Finally, problems with shared compartments (e.g., Coelho and Laporte, 2015; Yahyaoui et al., 2020) are also introduced, versus problems with unshared compartments (e.g., Brown et al., 1987; Hsu et al., 2007). In the context of perishable products, several papers are published (see, e.g., Chajakis and Guignard, 2003; Hsiao et al., 2017; Ostermeier and Hübner, 2018). For a comprehensive overview of state-of-the-art on heuristic and exact approaches for MCVRPs, the reader is referred to Ostermeier et al. (2021). Most solution approaches for the MCVRP are based on heuristics (see, e.g., Cornillier et al., 2008; Derigs et al., 2011; Mendoza et al., 2011; Cornillier et al., 2012). Compared to heuristics, only a few exact approaches are developed (see, e.g., Avella et al., 2004; Cornillier et al., 2008; Coelho and Laporte, 2015; Heßler, 2021; Heßler and Irnich, 2022). More recently, Aerts-Veenstra et al. (2024) developed a unified branch-price-and-cut algorithm for multi-compartment pickup and delivery problems. Their method accommodates several MCVRP variants and demonstrates flexibility and computational efficiency across diverse test instances, advancing the exact solution methods for complex vehicle routing scenarios involving compartmentalization.

In relation to VRPs with perishable products, we consider temperature-sensitive products. However, we do not embed a quality decay function in the objective function. Instead, we ensure product quality by imposing that cargo temperature respects the allowed temperature range of the consolidated products. In relation to VRPs with incompatibility constraints, we not only consider

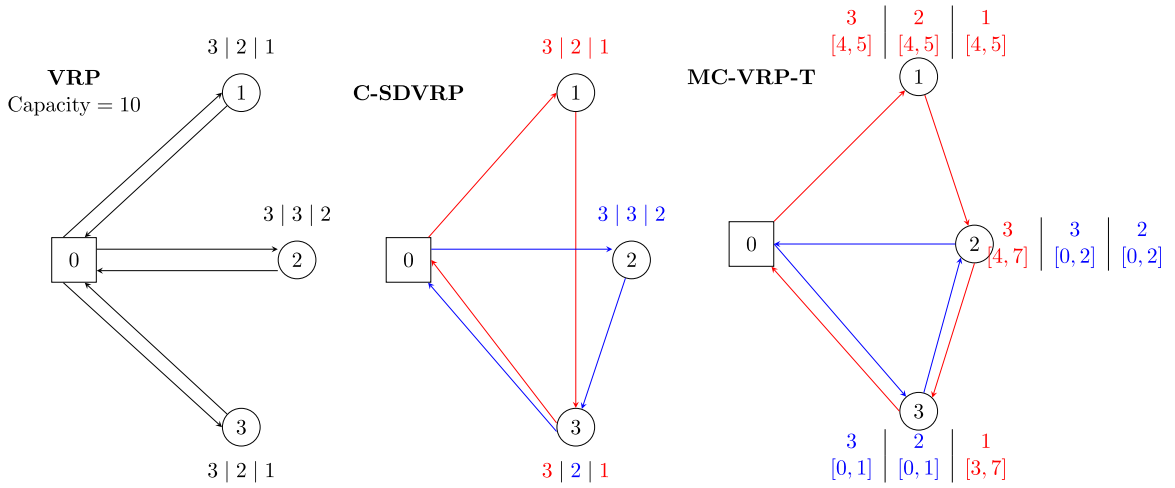


Fig. 1. Illustrative example based on Archetti et al. (2016).

incompatibility due to different storage and transportation temperatures as in Deng et al. (2022), but we also optimize vehicle target temperatures. This differs from MCVRPs with fixed compartments and preset temperatures, where no temperature decisions arise. In our single-compartment setting, temperature is a decision variable and feasible groupings follow from intersecting product ranges, so temperature selection integrates with routing and loading. With a fixed temperature, one route-level check suffices. With adjustable temperatures, a temperature is chosen on each arc and compatibility is verified per arc with the remaining load, coupling routing, flow, and temperature-dependent costs.

3. Problem description and formulation

Consider a set of geographically dispersed customers, each requesting a certain quantity of different products (commodities). Each commodity must be transported under certain temperature conditions and, similar to Archetti et al. (2015, 2016), each customer can be visited multiple times such that each visit delivers a different set of commodities. However, if a vehicle transports a specific commodity to a customer, then it must deliver the complete quantity of that commodity to the customer. In other words, it is not possible to split the delivery of individual commodities among different vehicles. A set of commodities can be assigned to the same vehicle if a target temperature exists that satisfies the temperature requirements of all the commodities in the set.

Given a homogeneous fleet of capacitated vehicles, the objective is to determine optimal vehicle routes and target temperatures that minimize the total cost, including routing and refrigeration costs-referred to as *temperature costs*-while satisfying both temperature and capacity constraints.

Fig. 1 illustrates the similarities and differences between VRP, the Commodity-Constrained Split Delivery VRP (C-SDVRP) studied in Archetti et al. (2015, 2016), and MC-VRP-T based on the example instance presented in Archetti et al. (2016) where each vehicle has a capacity of 10 units. The square represents the depot and circles the customers and the numbers above each customer denote the demand for each of the three commodities per customer. The colors blue and red are used to distinguish different vehicles.

In a VRP setting, the capacity restriction implies that each vehicle has to serve a single customer. In the C-SDVRP case from Archetti et al. (2016), the demand of customers 2 and 3 can be split, lowering the number of vehicles used and total costs. In the newly proposed MC-VRP-T, each commodity has a temperature range in which the product can be transported. For example, customer 2 requires three commodities: the first must be transported between 4 and 7° C and the second and third between 0 and 2 ° C. As a result, the solution presented for the C-SDVRP is no longer feasible because of the incompatibility caused by the temperature requirements. The same remark applies to customer 3 and the optimal solution consists of one vehicle operating at a target temperature between 4 and 5 ° C to deliver three commodities to customer 1, one commodity to customer 2 and a single commodity to customer 3. The remaining commodities are delivered by a second vehicle with a target temperature between 0 and 1 ° C.

The MC-VRP-T can be formalized on a complete directed graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$. The set $\mathcal{V} = \{0, 1, \dots, n\}$ represents the nodes, where node 0 denotes the starting and ending depot, and the set $\mathcal{V}' = \mathcal{V} \setminus \{0\}$ represents the customers. The set $A = \{(i, j) \in \mathcal{V} \times \mathcal{V} : i \neq j\}$ represents the arcs between all nodes. Each customer $i \in \mathcal{V}'$ requests a set of commodities $\mathcal{B}_i$. To simplify notation, the products requested by a customer receive a unique label, even when a product is requested by different customers, i.e., $\mathcal{B}_i \cap \mathcal{B}_j = \emptyset$, for each $i, j \in \mathcal{V}'$. We further denote $\mathcal{B} = \bigcup_{i \in \mathcal{V}} \mathcal{B}_i$ as the set of all the commodities. For each commodity $b \in \mathcal{B}$, $\left[\underline{T}_b, \overline{T}_b\right]$ represents its range of feasible transportation temperatures, and q_b represents its demand. Hence, the total demand of a customer $j \in \mathcal{V} \setminus \{0\}$ is $\sum_{b \in \mathcal{B}_j} q_b$. The set of vehicles is denoted as $\mathcal{K}$. Each vehicle has a capacity Q. We denote the external temperature during operations to be T.

When a vehicle traverses an edge $(i, j) \in \mathcal{E}$, two costs are incurred. First, the routing cost is denoted as c_{ij}. Second, the temperature control cost is caused by the cooling/heating needed to maintain the target temperature inside a truck. It is worth noting that temperature control (often refrigeration) is needed because of temperature changes in the cargo space during both loading and

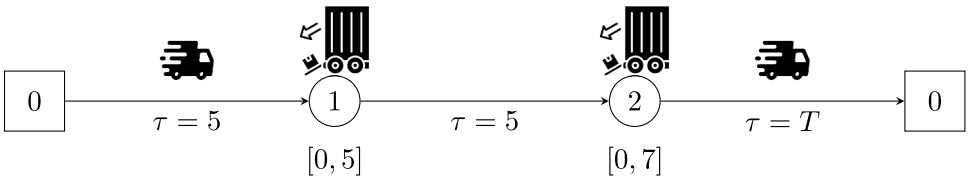


Fig. 2. Graphical representation of Example 1.

transport operations due to the external temperature. We refer to the absolute difference between the external temperature and the target temperature as the *temperature deviation*. The refrigeration cost caused by loading is calculated by multiplying the temperature deviation by a factor γ, while the refrigeration cost caused by transportation is calculated by multiplying the temperature deviation by a factor $\delta \times c_{ij}$.

The parameter δ represents the temperature-control cost per unit of travel duration, i.e., the cost of compensating for a temperature deviation during transportation. In our case study, the routing cost c_{ij} is proportional to the expected driving time, which allows us to express the transportation-related temperature cost as $\delta \times c_{ij}$. In line with common modeling practice for temperature-sensitive logistics Rong et al. (2011), Stellingwerf et al. (2021), δ is treated as a constant with units "cost per degree per unit of time."

The parameter γ models the temperature-control effort associated with loading and unloading operations, which induce heat exchange due to door openings and handling activities. This interpretation is consistent with the role of γ in quality-decay models (Song and Ko, 2016; Stellingwerf et al., 2021), where it captures stop-related thermal disturbances. Accordingly, γ has units "cost per degree" and is typically assumed constant when handling durations do not vary significantly across customers.

Without loss of generality, we assume that temperature control costs are only paid for edges $(i, j) \in \mathcal{E}$ for which $j \neq 0$, i.e., we assume that after visiting the last customer, the refrigeration units can be switched off while traveling to the depot. This structure of the objective function is inspired by the one presented in Stellingwerf et al. (2021) and is formulated in consultation with a leading logistics service provider specialized in temperature-controlled transport. Note, however, that our models and solution approaches in Section 4.2 can handle a wider class of temperature cost functions. In particular, if travel times t_{ij} or commodity-dependent handling times are available, the parameters δ and γ can be defined using these more detailed inputs without requiring structural changes to our formulations, and can capture richer temperature-cost structures analogous to those used in the literature (Rong et al., 2011; Stellingwerf et al., 2021).

The following example illustrates the objective function of the MC-VRP-T.

Example 1. Consider an instance of the MC-VRP-T in which $\mathcal{V} = \{0, 1, 2\}$. Each node requests one commodity, i.e., $\mathcal{B}_1 = \{1\}$ and $\mathcal{B}_2 = \{2\}$. The temperature requirements are the following: $\underline{T}_1 = \underline{T}_2 = 0$, $\overline{T}_1 = 5$, and $\overline{T}_2 = 7$. We assume the external temperature to be $T = 10$. In Fig. 2, we depict a feasible solution for this instance in which τ represents the optimal target temperature and calculate the corresponding total cost.

The vehicle starts from the depot, visits node 1 then node 2, before it drives back empty to the depot. The routing cost is therefore equal to $c_{01} + c_{12} + c_{20}$. We assume the truck drives with a cargo target temperature equal to 5° C (denoted in the figure by τ), which satisfies the temperature requirements of the two commodities. The corresponding temperature cost is given by $|T - 5| \times (\gamma + c_{01}\delta) + |T - 5| \times (\gamma + c_{12}\delta)$. We note that while traversing arc $(2, 0)$ the cooling units stop working because the truck is empty, hence setting τ equal to T. The first term represents the cost of maintaining a temperature of 5° C while travelling from node 0 to node 1 and unloading at node 1. The second component counts the temperature cost while driving from node 1 to node 2 and unloading at node 2.

A formulation for the MC-VRP-T can be obtained by extending the classical three-index vehicle-flow formulation used in Archetti et al. (2016) for the C-SDVRP. We use additional decision variables and constraints to model target temperature decisions and commodity temperature requirements. We define the formulation by introducing the following decision variables.

- $x_{ij}^k \in \{0, 1\}$: equal to 1 if vehicle $k \in \mathcal{K}$ traverses arc $(i, j) \in \mathcal{E}$, 0 otherwise.
- $z_i^k \in \{0, 1\}$: equal to 1 if node $i \in \mathcal{V}$ is visited by vehicle $k \in \mathcal{K}$, 0 otherwise.
- $y_{ib}^k \in \{0, 1\}$: equal to 1 if vehicle $k \in \mathcal{K}$ delivers commodity b to customer i, $i \in \mathcal{V}$, $b \in \mathcal{B}_i$, 0 otherwise.
- $\tau_k \in \mathbb{R}$: target temperature for vehicle $k \in \mathcal{K}$.
- α_{ij}^k: absolute deviation between external temperature T and the target temperature τ_k of vehicle $k \in \mathcal{K}$ (i.e., $|T - \tau_k|$) while traversing edge $(i, j) \in \mathcal{E}$, $j \neq 0$. Note that this variable is not defined for $j = 0$ because we assume that cooling units are switched off on edges $(i, 0)$.

The three-index vehicle-flow formulation of the MC-VRP-T is as follows:

$$\min \sum_{k \in \mathcal{K}} \left(\sum_{(i,j) \in \mathcal{E}} c_{ij} x_{ij}^k + \sum_{(i,j) \in \mathcal{E} : j \neq 0} (\gamma + \delta c_{ij}) \alpha_{ij}^k \right) \tag{1a}$$

$$\text{s.t.} \sum_{k \in \mathcal{K}} y_{ib}^k = 1 \qquad \forall i \in \mathcal{V}', \forall b \in \mathcal{B}_i \tag{1b}$$

$$\sum_{(i,j) \in \mathcal{E}} x_{ij}^k + \sum_{(j,i) \in \mathcal{E}} x_{ij}^k = 2 z_{ik} \qquad \forall i \in \mathcal{V}, \forall k \in \mathcal{K} \tag{1c}$$

$$\sum_{(0,i)\in\mathcal{E}} x_{0i}^k = \sum_{(i,0)\in\mathcal{E}} x_{i0}^k \qquad \forall i \in \mathcal{V}', \forall k \in \mathcal{K} \tag{1d}$$

$$y_{ib}^k \le z_{ik} \qquad \forall i \in \mathcal{V}', \forall k \in \mathcal{K}, \forall b \in \mathcal{B}_i \tag{1e}$$

$$\sum_{i\in\mathcal{V}} \sum_{b\in\mathcal{B}_i} q_b y_{ib}^k \le Q z_{0k} \qquad \forall k \in \mathcal{K} \tag{1f}$$

$$\sum_{i\in S} \sum_{j\in S: j<i} x_{ij}^k \le \sum_{i\in S} z_{ik} - z_{uk} \qquad \forall S \subseteq \mathcal{V}', u \in S, k \in \mathcal{K} \tag{1g}$$

$$\tau_k \le \overline{T}_b + (1 - y_{ib}^k)(M - \overline{T}_b) \qquad \forall i \in \mathcal{V}', \forall b \in \mathcal{B}_i, \forall k \in \mathcal{K} \tag{1h}$$

$$\tau_k \ge \underline{T}_b - (1 - y_{ib}^k)(M + \underline{T}_b) \qquad \forall i \in \mathcal{V}', \forall b \in \mathcal{B}_i, \forall k \in \mathcal{K} \tag{1i}$$

$$\alpha_{ij}^k \ge T - \tau_k - \left(1 - x_{ij}^k\right) M \qquad \forall (i,j) \in \mathcal{E}, j \ne 0, \forall k \in \mathcal{K} \tag{1j}$$

$$\alpha_{ij}^k \ge \tau_k - T - \left(1 - x_{ij}^k\right) M \qquad \forall (i,j) \in \mathcal{E}, j \ne 0, \forall k \in \mathcal{K} \tag{1k}$$

$$x_{ij}^k \in \{0,1\} \qquad \forall (i,j) \in \mathcal{E}, \forall k \in \mathcal{K} \tag{1l}$$

$$z_{ik} \in \{0,1\} \qquad \forall i \in \mathcal{V}, \forall k \in \mathcal{K} \tag{1m}$$

$$y_{ib}^k \in \{0,1\} \qquad \forall i \in \mathcal{V}', \forall k \in \mathcal{K}, \forall b \in \mathcal{B}_i \tag{1n}$$

$$\tau_k \in \mathbb{R} \qquad \forall k \in \mathcal{K} \tag{1o}$$

$$\alpha_{ij}^k \ge 0 \qquad \forall (i,j) \in \mathcal{E}, j \ne 0, \forall k \in \mathcal{K} \tag{1p}$$

The objective function (1a) consists of travel and temperature-dependent costs during loading and transport operations. Constraints (1b) ensure that the delivery of a commodity requested by a customer can not be split, i.e., it is served by a single vehicle. Constraints (1c) are the degree constraints, while constraints (1d) ensure flow conservation at the depot. Constraints (1e) make sure a commodity requested by a customer can be delivered by a certain vehicle only if the vehicle visits the customer. Constraints (1f) ensure that the vehicle capacity is not exceeded and sub-tours are prevented by constraints (1g). Constraints (1h) and (1i) set the target temperature of each vehicle. A valid big-M in these constraints is $M = \max\left\{|T|, |\min_{i\in\mathcal{B}}\left\{\underline{T}_i\right\}|, |\max_{i\in\mathcal{B}}\left\{\overline{T}_i\right\}|\right\}$. Constraints (1j) – (1k) are used to set the absolute target temperature deviation and to link variables α_{ij}^k, τ_k, and x_{ij}^k for each $(i,j) \in \mathcal{E}$, $j \ne 0$, and $k \in \mathcal{K}$. Constraints (1l) – (1p) define the domain of the decision variables.

4. Problem reformulations

The relaxation of formulation (1) is expected to provide a weak lower bound because of the many big-*M* constraints it contains. Hence, this section studies the structure of the proposed MC-VRP-T and presents alternative, stronger formulations. Moreover, we transform the MC-VRP-T into a *Heterogenous Fleet Capacitated Vehicle Routing Problem* (HFVRP) (Baldacci et al., 2008) that can be solved using the generic branch-cut-and-price (BPC) solver proposed in Pessoa et al. (2020), Sadykov and Vanderbeck (2021).

To this end, section 4.1 presents a key property of our problem that will be used to derive the reformulations presented in Section 4.2 and 4.3. This property is presented in Theorem 1.

4.1. Optimal temperature set

Theorem 1. *There is at least one optimal solution of problem (1) for which the optimal target temperatures τ_k^*, $k \in \mathcal{K}$, belong to the finite set*

$$\mathcal{T} = \left\{ \bigcup_{b\in\mathcal{B}} \{\underline{T}_b, \overline{T}_b\} \cup \{T\} \right\}.$$

Proof. We prove the theorem by contradiction. Assume that we have a feasible optimal solution of problem (1) in which $\tau_k \notin \mathcal{T}$ for a certain $k \in \mathcal{K}$. Let $\tilde{\mathcal{B}} \subseteq \mathcal{B}$ be the subset of the commodities transported by vehicle k. Since the solution is feasible and $\tau_k \notin \mathcal{T}$, the following must hold:

$$\max_{b\in\tilde{\mathcal{B}}}\{\underline{T}_b\} < \tau_k < \min_{b\in\tilde{\mathcal{B}}}\{\overline{T}_b\}. \tag{2}$$

Since the temperature-dependent costs are monotonically increasing with respect to the deviation of τ_k from T, a better solution can be obtained by:

- setting $\tau_k = T$ if $T \in \left[\max_{b\in\tilde{\mathcal{B}}}\{\underline{T}_b\}, \min_{b\in\tilde{\mathcal{B}}}\{\overline{T}_b\}\right]$,
- setting $\tau_k = \max_{b\in\tilde{\mathcal{B}}}\{\underline{T}_b\}$ if $T \le \max_{b\in\tilde{\mathcal{B}}}\{\underline{T}_b\}$,
- setting $\tau_k = \min_{b\in\tilde{\mathcal{B}}}\{\overline{T}_b\}$ if $T \ge \min_{b\in\tilde{\mathcal{B}}}\{\overline{T}_b\}$.

Thus, τ_k must belong to $\mathcal{T}$. □

Theorem 1 holds for a wide class of temperature cost functions, beyond the cost structure used in formulation (1). The only assumption used in the proof is that the temperature cost is monotonically non-decreasing with respect to $|T - \tau_k|$ for each $k \in \mathcal{K}$. Therefore, Theorem 1 applies to any version of the MC-VRP-T in which the adopted temperature cost function satisfies this condition, such as those in Stellingwerf et al. (2018, 2021), Lin et al. (2025).

4.2. Temperature-flow formulation

Based on Theorem 1, the search for the optimal value of $\{\tau_k\}_{k\in\mathcal{K}}$ can be limited to the set $\mathcal{T} = \left\{\bigcup_{b\in\mathcal{B}} \left\{\underline{T}_b, \overline{T}_b\right\} \cup T\right\}$. We use this property to propose an improved formulation for the MC-VRP-T, which we name *temperature-flow formulation*.

The temperature-flow formulation is defined on an expanded directed graph $\widetilde{\mathcal{G}}(\mathcal{N}, \mathcal{A})$ where $\mathcal{N} = \bigcup_{i\in\mathcal{V}} \mathcal{B}_i \bigcup \{0\}$ represents the set of nodes, obtained by duplicating the original customer nodes for each requested commodity. The set of arcs $\mathcal{A} = \left\{(i,j), i,j \in \mathcal{N}, i \neq j : \left[\underline{T}_i, \overline{T}_i\right] \bigcap \left[\underline{T}_j, \overline{T}_j\right] \neq \emptyset\right\} \cup \left\{(i,0), i \in \mathcal{N} \setminus \{0\}\right\}$ does not include arcs connecting incompatible pairs of commodities and therefore the graph $\widetilde{\mathcal{G}}(\mathcal{N}, \mathcal{A})$ is sparse. For each temperature $t \in \mathcal{T}$, we define a sub-graph $\widetilde{\mathcal{G}}_t(\mathcal{N}_t, \mathcal{A}_t)$ of graph $\widetilde{\mathcal{G}}(\mathcal{N}, \mathcal{A})$, where $\mathcal{N}_t = \left\{i \in \mathcal{N} : t \in \left[\underline{T}_i, \overline{T}_i\right]\right\}$ contains only nodes (commodities) compatible with temperature t and $\mathcal{A}_t = \left\{(i,j) \in \mathcal{A} : i,j \in \mathcal{N}_t\right\}$ is the corresponding set of arcs. We denote with $\mathcal{N}'$ and $\mathcal{N}'_t$ sets $\mathcal{N} \setminus \{0\}$ and $\mathcal{N}_t \setminus \{0\}, t \in \mathcal{T}$, respectively.

We denote the total cost (routing plus temperature) for traversing arc $(i,j) \in \mathcal{A}_t$ at temperature $t \in \mathcal{T}$ as c^t_{ij}. If i and j represent deliveries of different commodities to the same customer in graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$ (i.e., the original graph), then $c^t_{ij} = 0$ for each $t \in \mathcal{T}$.

Theorem 1 allows us to formulate an alternative model that is equivalent to model (1) for the MC-VRP-T using decision variables $x^t_{ij} \in \{0,1\}$ that take value 1 if a vehicle with target temperature $t \in \mathcal{T}$ traverses arc $(i,j) \in \mathcal{A}_t$, 0 otherwise, and variables ℓ^b_{ij}, which represent the flow of commodity $b \in \mathcal{B}$ on arc $(i,j) \in \mathcal{A}$. The temperature-flow formulation states as follows:

$$\min \sum_{(i,j)\in\mathcal{A}} \sum_{t\in\mathcal{T}} c^t_{ij} x^t_{ij} \tag{3a}$$

$$\text{s.t.} \sum_{t\in\mathcal{T}} \sum_{(i,j)\in\mathcal{A}_t} x^t_{ij} = 1 \qquad \forall i \in \mathcal{N}' \tag{3b}$$

$$\sum_{(i,j)\in\mathcal{A}_t} x^t_{ij} = \sum_{(j,i)\in\mathcal{A}_t} x^t_{ji} \qquad \forall i \in \mathcal{N}, \forall t \in \mathcal{T} \tag{3c}$$

$$\sum_{t\in\mathcal{T}} \sum_{(0,j)\in\mathcal{A}_t} x^t_{0j} \leq |\mathcal{K}| \tag{3d}$$

$$\sum_{(j,i)\in\mathcal{A}} \ell^b_{ji} - \sum_{(i,j)\in\mathcal{A}} \ell^b_{ij} = \mathbb{1}\{i = b\} q_b \qquad \forall i, b \in \mathcal{N}', \tag{3e}$$

$$\ell^b_{ij} \leq \sum_{t\in\mathcal{T}: t\in[\underline{T}_b, \overline{T}_b]} q_b x^t_{ij} \qquad \forall b \in \mathcal{N} \setminus \{0\}, \forall (i,j) \in \mathcal{A}, \tag{3f}$$

$$\sum_{b\in\mathcal{N}'} \ell^b_{ij} \leq Q - q_j \qquad \forall (i,j) \in \mathcal{A}. \tag{3g}$$

$$x^t_{ij} \in \{0,1\} \qquad \forall t \in \mathcal{T}, \forall (i,j) \in \mathcal{A} \tag{3h}$$

$$\ell^b_{ij} \in \mathbb{R}_+ \qquad \forall b \in \mathcal{N}', \forall (i,j) \in \mathcal{A} \tag{3i}$$

Objective function (3a) minimizes the total cost. Constraints (3b) ensures all the commodities are delivered. Constraints (3c) enforce flow conservation at each node for each temperature and constraint (3d) ensures the maximum number of vehicles is not exceeded. We define $\mathbb{1}\{i = b\}$ to be the indicator function that takes value 1 if $i = b$ and 0 otherwise. Constraints (3e) ensure flow conservation for each commodity at each node. Constraints (3f) link routing and commodity-flow variables and enforce temperature requirements on the flow-variable. Finally, constraints (3g) make sure the vehicle capacity is not exceeded. The domain of the decision variables is defined by constraints (3h) and (3i).

Note that the temperature-flow formulation can incorporate more complex temperature cost structures, such as ones considering service-time-dependent temperature losses and higher heat infiltration during handling operations. Assuming a deterministic service time at each customer and a given function to quantify the loss, we could use this information to adjust the cost parameters $\{c^t_{ij}\}_{t\in\mathcal{T},(i,j)\in\mathcal{A}}$ to reflect the increased heat gain during handling at customer i. This would allow us to capture the effect of service-time-dependent temperature loss without changing the structure of the model.

4.2.1. Valid inequalities

To strengthen the linear relaxation of formulation (3), we include the following valid inequalities.

We introduce the *Minimum Number of Vehicles* inequalities. They have the following form:

$$\sum_{(0,i)\in\mathcal{A}_t} Q x^t_{0i} \geq \sum_{(i,j)\in\mathcal{A}_t, i\neq 0} q_i x^t_{ij} \quad \forall t \in \mathcal{T}. \tag{4}$$

Inequalities (4) stipulate that for each temperature, $t \in \mathcal{T}$, the sum of the total vehicle capacity (i.e., number of used vehicles times Q) should be greater or equal to the demand of commodities transported at temperature t.

Finally, we introduce the *Temperature-Aggregated Rounded Capacity* Cuts, which have the form:

$$\sum_{t\in\mathcal{T}}\sum_{(i,j)\in\mathcal{A}_t:i,j\in S} x_{ij}^t \le |S| - \left\lceil \frac{\sum_{i\in S} q_i}{Q} \right\rceil \quad \forall S \subseteq \mathcal{N}', |S| \ge 2. \tag{5}$$

They are obtained by aggregating the rounded capacity cuts, that can be defined for each temperature $t \in \mathcal{T}$ as follows:

$$\sum_{(i,j)\in\mathcal{A}_t:i,j\in S} x_{ij}^t \le |S| - \left\lceil \frac{\sum_{i\in S} q_i}{Q} \right\rceil \quad \forall S \subseteq \mathcal{N}', |S| \ge 2, \forall t \in \mathcal{T}. \tag{6}$$

Because (5) are obtained by aggregating (6) over the set of temperatures $\mathcal{T}$, they are valid for our problem. As a consequence, we try to separate them first before we proceed with the separation of inequalities (6) when none are violated. Further details of our branch-and-cut algorithm are provided in the next section.

4.2.2. Branch & cut algorithm

We solve formulation (3) by a branch-and-cut algorithm to include the exponentially many (aggregated) rounded capacity cuts. Inequalities (4) are all added to formulation at the beginning while (6) and (5) are added dynamically as it is explained below.

At each node of the branch-and-bound tree, our cutting scheme first looks for violated aggregated-temperature rounded capacity cuts (5).

If a violated cut is found, then we use the corresponding set $S \subseteq \mathcal{N}'$ to find potentially violated (disaggregated) rounded capacity cuts (6) for each temperature $t \in \mathcal{T}$. In other words, we derive a new cut (6) for each $t \in \mathcal{T}$ defined by the subset S containing the nodes that can be served at temperature t, i.e., $\mathcal{N}_t' \cap S$. Every violated disaggregated cut is added to the current linear relaxation. If no violated disaggregated cuts are found, we add the violated temperature-aggregate rounded capacity cut.

If no violated temperature-aggregated rounded capacity cuts are detected, the algorithm proceeds with the separation of inequalities (6) for each $t \in \mathcal{T}$. When a violated cut is found for a given temperature t, we check whether the cut is violated for other temperatures in the set $\mathcal{T}$. The procedure is repeated every time an integer solution is encountered and for every fractional solution at each node of the branch-and-bound tree. When no more cuts are detected, the algorithm resorts to branching. All the separation problems are solved as described in Lysgaard (2003). In particular, we create a supporting graph for the current fractional solution containing all the arcs with a positive flow in the current LP solution. Each arc has a weight equal to the fractional value of the corresponding variable. If we are looking for violated temperature-aggregated rounded capacity cuts, we create a single supporting graph containing all the arcs with positive flow across all temperatures. If we are looking for violated disaggregated rounded capacity cuts, we create a supporting graph for each temperature $t \in \mathcal{T}$ containing only the arcs with positive flow at temperature t. Using this graph, we apply heuristic procedures as described in Lysgaard (2003) using the CVRPSEP package (Lysgaard, 2003). This is embedded in the callback of the branch-and-cut algorithm of the solver, and the separation is performed at every node of the search tree.

4.3. Transformation to heterogeneous fleet vehicle routing problem

In the HFVRP, a fleet of heterogeneous vehicles is available to serve customers requesting a certain amount of product. Each vehicle belongs to a certain type, and, the routing cost for traversing an edge depends on the type of vehicle used. The goal is to define which vehicle types to use and define their routes to visit all the customers at the minimum cost. In this section, we show how the MC-VRP-T can be cast as an HFVRP leveraging Theorem 1. We use the same notation introduced in Section 4.2.

We define the HFVRP on the extended graph $\mathcal{G}(\mathcal{N}, \mathcal{A})$. The temperature set $\mathcal{T}$ encodes the vehicle types. The cost of traversing an arc $(i, j) \in \mathcal{A}$ is equal to c_{ij}^t if $(i, j) \in \mathcal{A}_t$ and ∞ otherwise. The total number of used vehicles must not exceed $|\mathcal{K}|$. Let $\mathcal{R}_t$ be the set of all the elementary feasible routes that can be performed by vehicle type (or at temperature) $t \in \mathcal{T}$. Accordingly, the pricing step consists of solving a separate resource-constrained shortest path problem for each $t \in \mathcal{T}$, each defined on its own temperature-specific graph and cost structure. The total cost (routing plus temperature) for selecting a route $r \in \mathcal{R}_t, t \in \mathcal{T}$ is equal to p_r. Let a_{ir} be a binary coefficient equal to 1 if route $r \in \mathcal{R}_t, t \in \mathcal{T}$ serves customer $i \in \mathcal{N}_t'$. Using variables λ_r equal to 1 if route $r \in \mathcal{R}_t,\ t \in \mathcal{T}$, the set partitioning formulation of the HFVRP states as follows.

$$\min \sum_{t\in\mathcal{T}}\sum_{r\in\mathcal{R}_t} p_r\lambda_r \tag{7a}$$

$$\text{s.t.} \sum_{t\in\mathcal{T}:i\in\mathcal{N}_t'}\sum_{r\in\mathcal{R}_t} a_{ir}\lambda_r = 1 \qquad \forall i \in \mathcal{N}' \tag{7b}$$

$$\sum_{t\in\mathcal{T}}\sum_{r\in\mathcal{R}_t} \lambda_r \le |\mathcal{K}| \tag{7c}$$

$$\lambda_r \in \{0, 1\} \qquad \forall t \in \mathcal{T}, \forall r \in \mathcal{R}_t \tag{7d}$$

To solve problem (7), we adopt the BPC general solver introduced in Pessoa et al. (2020), Sadykov and Vanderbeck (2021). We refer the reader to Pessoa et al. (2020) for all details on the BCP algorithm (e.g., relaxed master problem, pricing problem, cuts, acceleration techniques).

Having established our formulations and solution methods for the constant-temperature variant, we now turn to a more flexible setting in which the vehicle's target temperature may be adjusted after each stop.

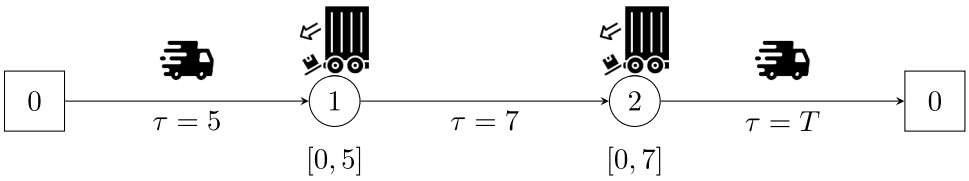


Fig. 3. Graphical representation of Example 2.

5. The MC-VRP-T with adjustable temperature

In this section, we introduce and model an extension of the MC-VRP-T in which the target temperature of each vehicle can be adjusted after each stop. Section 5.1 describes the problem and Section 5.2 defines a mathematical formulation.

5.1. Description and motivation

In a feasible solution of the MC-VRP-T, the target temperature of each vehicle is determined based on the temperature requirements of all transported commodities. The target temperature is set at the beginning of the route and maintained until the last customer of the route is visited as described in Section 3. However, after having visited a customer, the set of commodities transported by the vehicle is reduced, and therefore the target temperature could be adjusted, accounting only for the temperature requirement of the remaining commodities. This temperature adjustment also shows clear analogies with other contexts in the literature. In production planning, machines adjustments between product families typically require a setup operation that incurs time or cost, which is conceptually similar to adjusting a temperature level along a route (Fanjul-Peyro et al., 2019). A related analogy arises in the pollution routing problem, where the vehicle's travel speed is a decision affecting fuel consumption and emissions; changing speed on consecutive arcs plays a role similar to modifying the temperature on different legs of a route (e.g., Demir et al. (2014)). The following example illustrates the benefits of adjusting target temperatures along the route. We use the same instance as for Example 1.

Example 2. If we are allowed to adjust the temperature after visiting customer 1 (and dropping commodity 1), it will be cheaper to increase the target temperature from 5° C to 7° C on arc $(1, 2)$. This solution is illustrated in Fig. 3.

The temperature cost of this solution is equal to $|T - 5| \times (\gamma + c_{01}\delta) + |T - 7| \times (\gamma + c_{12}\delta)$, which results in a saving of $2(\gamma + c_{12}\delta)$ compared to the solution in Example 1 given that $T = 10$.

Although allowing for temperature adjustments during the route can reduce costs, this approach is not always practically desirable, especially when drivers must reset the temperature manually, which may lead to mistakes and consequently deteriorated product quality, as pointed out by our industry partner. Moreover, additional IT infrastructure may be required to integrate routing and temperature decisions. Nevertheless, it is worth studying the MC-VRP-T with Adjustable Temperature (MC-VRP-TA) to shed light on the benefits that may be achieved in terms of costs, and the impact on solution structure and complexity. The derived insights can help companies make decisions related to future investments in IT integration and automation. We note that if costs associated with changing temperature levels must also be considered, this can be modeled by introducing explicit temperature adjustment costs, as discussed in Section 5.2.

5.2. Mathematical formulation

Theorem 1 holds for the MC-VRP-TA. The optimal temperature set $\mathcal{T}$ defined for the MC-VRP-T, also guarantees optimality for the MC-VRP-TA. The proof follows the same arguments used to prove Theorem 1, repeated for each arc of the route of every vehicle since a new temperature can be set after each stop. Therefore, we can obtain a formulation for MC-VRP-TA by slightly modifying the temperature-flow formulation introduced in Section 4.2. We use the same notation as for formulation (3) and additionally define the sets of arcs $\mathcal{A}'_t$ defined for each $t \in \mathcal{T}$. For a temperature $t \in \mathcal{T}$, the set $\mathcal{A}'_t$ contains all the arc $(i, j) \in \mathcal{A}$ such that (a) the connected nodes i, j have at least one common feasible temperature and (b) the commodity corresponding to node j can be transported at temperature t, i.e.,

$$\mathcal{A}'_t := \left\{ (i,j) \in \mathcal{A} : t \in [\underline{T}_i, \overline{T}_i],\ [\underline{T}_i, \overline{T}_i] \cup [\underline{T}_j, \overline{T}_j] \neq \emptyset \right\}.$$

The temperature-flow formulation for the MC-VRP-TA is as follows:

$$\min \sum_{t \in \mathcal{T}} \sum_{(i,j) \in \mathcal{A}'_t} c^t_{ij} x^t_{ij} \tag{8a}$$

$$\text{s.t.} \sum_{t \in \mathcal{T}} \sum_{(i,j) \in \mathcal{A}'_t} x^t_{ij} = 1 \qquad \forall i \in \mathcal{N}', \forall t \in \mathcal{T} \tag{8b}$$

$$\sum_{t \in \mathcal{T}} \sum_{(i,j) \in \mathcal{A}'_t} x^t_{ij} = \sum_{t \in \mathcal{T}} \sum_{(j,i) \in \mathcal{A}'_t} x^t_{ji} \qquad \forall i \in \mathcal{N} \tag{8c}$$

$$\sum_{t\in\mathcal{T}}\sum_{(0,j)\in\mathcal{A}'_t} x^t_{0j} \le |\mathcal{K}| \tag{8d}$$

$$\sum_{(j,i)\in\mathcal{A}} \ell^b_{ji} - \sum_{(i,j)\in\mathcal{A}} \ell^b_{ij} = \mathbb{1}\{i=b\}q_b \qquad \forall i,b\in\mathcal{N}' \tag{8e}$$

$$\ell^b_{ij} \le \sum_{t\in\mathcal{T}:t\in[\underline{T}_b,\overline{T}_b]} x^t_{ij} q_b \qquad \forall b\in\tilde{\mathcal{N}},\forall (i,j)\in\mathcal{A} \tag{8f}$$

$$x^t_{ij}\in\{0,1\} \qquad \forall t\in\mathcal{T},\forall (i,j)\in\mathcal{A}'_t \tag{8g}$$

$$\ell^b_{ij}\in\mathbb{R}_+ \qquad \forall b\in\mathcal{N}',\forall (i,j)\in\mathcal{A} \tag{8h}$$

The objective function (8a) minimizes the total cost while constraints (8b) – (8d) have the same role as constraints (3b) – (3d), i.e., they ensure all the nodes are visited once, guarantee flow-conservation, and make sure the maximum number of used vehicles and vehicle capacity are respected. Constraints (8e) – (8f) keep track of the commodities transported and ensure the temperatures set on each arc along a route satisfy the temperature requirements of the current cargo composition. Finally, constraints (8g) and (8h) define the variable domains.

Having defined our formulation for the MC-VRP-TA, we now discuss the differences with the formulation for the MC-VRP-T, some implications on the valid inequalities, and how formulation (8) can be adapted to incorporate costs for adjusting temperatures along the route.

MC-VRP-TA vs. MC-VRP-T Problem Structure. The structure of formulation (8) for the MC-VRP-TA exhibits some fundamental differences with the one of formulation (3).

In the MC-VRP-T, each route is assigned a single temperature, which must be maintained for all arcs traversed by the vehicle. This restriction allows the problem to be formulated with flow conservation defined separately for each temperature, resulting in a more decomposable structure. In contrast, the adjustable-temperature variant (MC-VRP-TA) permits temperature changes between arcs within the same route. This flexibility requires these constraints to be enforced over the entire graph, leading to a more integrated formulation. While the commodity-incompatibility structure remains present, since all products on a route must still be compatible with the initial temperature, the flexibility in temperature changes leads to more complex interactions and a larger search space.

A first consequence of this structural difference between formulations (3) and (8) lies in the number of required variables. The binary variables x^t_{ij} associated with each temperature $t\in\mathcal{T}$ are defined for each $(i,j)\in\mathcal{A}_t$ in formulation (3), while formulation (8) requires a variable for each arc (i,j) in set $\mathcal{A}'_t$ that is a superset of $\mathcal{A}_t$.

Another important difference is that the MC-VRP-TA cannot be converted into an HFVRP as we did for the MC-VRP-T in Section 4.3. Otherwise, this would mean the same route could potentially involve multiple types of vehicles. A set partitioning formulation similar to (7) could still be used as a master problem in a branch-and-price-and-cut framework by generating routes (columns) in which the temperature can change after each stop. However, the corresponding pricing problem has to incorporate temperature decisions and it is therefore more complex than the one that is solved for the HFVRP. In particular, temperature transitions can be handled directly during label extension in the pricing problem: each label must track the current temperature as an additional resource, and any change of temperature can be accounted for by adding the corresponding transition cost. For this reason, we only use the temperature-flow formulation for the MC-VRP-TA.

Temperature-Aggregated Rounded Capacity Cuts. Inequalities (5) are valid for formulation (8) as well because routes using different temperatures that violate (5) are infeasible for the MC-VRP-TA. We can therefore use them in a branch-and-cut algorithm to solve formulation (8) as we do for formulation (3). In contrast, this is not the case for the classical rounded capacity cuts (6) because they are defined for each temperature separately. In particular, a route that violates (6) for a given temperature t may still be feasible for the MC-VRP-TA if the vehicle can adjust its temperature to serve the customers in the route at different temperatures.

Cost of Adjusting Temperatures. Adjusting temperatures along the route may require additional costs, for example, due to the need for manual adjustments by drivers or the use of IT infrastructure. If these costs are relevant, they can be considered into formulation (8) incorporating additional decision variables and constraints. For instance, if we assume a fixed cost f for changing the temperature from any t to any $t', t\neq t', t,t'\in\mathcal{T}$, we can introduce binary variables $y_i, i\in\mathcal{N}'$ that indicate whether the temperature is changed at node i. The objective function can then be modified as follows:

$$\min \sum_{t\in\mathcal{T}}\sum_{(i,j)\in\mathcal{A}'_t} c^t_{ij}x^t_{ij} + f\sum_{i\in\mathcal{N}'} y_i, \tag{9}$$

where the second term accounts for the cost of changing the temperature at the nodes. To ensure that $\{y_i\}_{i\in\mathcal{N}'}$ are linked to the $\{x^t_{ij}\}_{t\in\mathcal{T},(i,j)\in\mathcal{A}'_t}$ variables, the following set of constraints must be added to the formulation:

$$y_i \ge \sum_{(j,i)\in\mathcal{A}'_t} x^t_{ji} - \sum_{(i,j)\in\mathcal{A}'_t} x^t_{ij} \quad \forall i\in\mathcal{N}',\forall t\in\mathcal{T},\forall (i,j)\in\mathcal{A}'_t \tag{10a}$$

$$y_i\in\{0,1\} \qquad \forall i\in\mathcal{N}'. \tag{10b}$$

Constraints (10a) can be seen as flow-conservation constraints for every temperature $t\in\mathcal{T}$, ensuring that if the flow of temperature t into node i differs from the flow of temperature t out of node i, then y_i must be set to 1, indicating a temperature change at

Table 1
Pool of commodities used to generate the dataset. For each commodity (Commodity) we report the minimum ($\underline{T}$) and maximum ($\overline{T}$) feasible transport temperature (° C).

Commodity	$\underline{T}$(° C)	$\overline{T}$(° C)	Commodity	$\underline{T}$(° C)	$\overline{T}$(° C)
1	3	6	7	0	2
2	13	16	8	5	10
3	7	10	9	3	7
4	11	13	10	0	4
5	10	13	11	0	1
6	4	7	12	13	21

node i. Constraints (10b) define the variable domain. This modelling approach can be extended to account for temperature-specific adjustment costs by introducing additional binary variables and constraints.

6. Computational experiments

In this section, we first describe how we developed problem instances for the MC-VRP-T (Section 6.1). In Section 6.2, we study the performance of alternative model formulations for the MC-VRP-T: the vehicle-flow formulation (1), the temperature-flow formulation (3), and the BCP solver for the HFVRP transformation of the problem discussed in Section 4.3. In the same section, we also discuss the impact of the valid inequalities introduced in Section 4.2.1 for the temperature-flow formulation (3). Section 6.3 reports on similar experiments for the single-commodity case. Section 6.4 analyses the additional benefits of allowing for adjustable target temperatures by comparing the solutions of the MC-VRP-T with the ones of the MC-VRP-TA. In Section 6.5 we use the MC-VRP-T model formulations to demonstrate the benefit of integrating routing and temperature decisions. All of our experiments are performed in the single core on a machine equipped with a 2.70GHz Intel i7-1265U processor and with 16GB of RAM. Formulations (1), (3), and (8) are solved using Gurobi 10.0.3 and its callback to add cuts dynamically. The BCP solver uses CPLEX 20.1 to solve the linear relaxations because it does not support Gurobi. All the models are implemented in C++. The time limit for each experiment is set to 3600 seconds.

6.1. Dataset

Because we are the first to define the MC-VRP-T no benchmark instances are available in the literature. Following Archetti et al. (2016), we therefore generate our datasets by combining the well-known Solomon instances (Solomon, 1987) and real data provided by the company Mandersloot, a logistics service provider specialized in temperature-sensitive products. The customer locations, the capacity of each vehicle, and the total demand of each customer are taken from the Solomon R101, C101, and RC101 instances. Then, 2 or 3 commodities from Table 1 are randomly assigned to each customer. The total demand of the customer (from the Solomon instances) is randomly split between the assigned commodities. The number of available vehicles is determined by solving a bin packing problem that packs commodities into bins (vehicles), considering temperature requirements and capacity to ensure feasibility. This study considers the first 5, 10, 15, 20, 25, 30, 35, and 40 customers for each Solomon network type (C, R, and RC), resulting in instances with an expanded graph ranging from 12 to 101 nodes. For each problem, we consider the cases of no temperature cost ($\gamma = 0$, $\delta = 0$), low/medium temperature cost ($\gamma = 0.5$, $\delta = 0.05$), and high temperature cost ($\gamma = 1$, $\delta = 0.1$). The external temperature T is set to 20° C. Additionally, to showcase the impact of the number of commodities, we generate a single-commodity dataset (i.e., every customer requests only one commodity) following the same procedure discussed above. For these problems, we show that our formulation can handle instances with up to 100 customers. All the instances can be found here.

6.2. Performance comparison on multi-commodity problems

In this section, we study the performance of the proposed formulations for the MC-VRP-T on the multi-commodity problems. Table 2 reports a summary of the results grouped per size of the network ($|\mathcal{V}'|/|\mathcal{B}|$) and per values of γ, δ (γ/δ). We denote with "VFF" the vehicle-flow formulation (1), with "TFF" the temperature-flow formulation (3) with the valid inequalities introduced in Section 4.2.1, and with "BCP" the formulation and algorithm discussed in Section 4.3. For each approach, we report the number of instances solved "Opt (#)", the average optimality gap of the instances not solved to optimality "Gap (%)", and the average computational time of the problems solved in seconds "Time (s)". When all the instances of a particular class are solved, "-" is reported under the Gap (%) columns. If a "x" is displayed, it means that the method was not able to find upper bounds (i.e., integer feasible solutions) within the time limit.

Consistent with previous VRP research, Table 2 demonstrates that VFF can only consistently solve instances up to 5 customers generating a network of up to 15 nodes. The BCP is more effective, consistently solving instances with up to 15 customers (corresponding to up to 40 nodes in the network), but it struggles to find feasible solutions when the size of the network grows. The merit of the temperature-flow formulation TFF, is illustrated by solving 69/72 instances involving up to 40 customers and 101 nodes. Moreover, for instances that are also solved by other methods, the temperature-flow formulation is consistently faster. This performance advantage is largely due to embedding temperature feasibility directly in the flow variables, which avoids big-M linking constraints and allows many infeasible arc–temperature combinations to be eliminated a priori.

Table 2
Performance of the vehicle-flow and temperature-flow formulations and of the BCP algorithm on multi-commodity problems.

		Opt (#)			Gap (%)			Time (s)		
$\lvert\mathcal{V}'\rvert/\lvert\mathcal{B}\rvert$	γ/δ	VFF	TFF	BCP	VFF	TFF	BCP	VFF	TFF	BCP
5/12–15	0/0	3/3	3/3	3/3	-	-	-	0.2	0.1	0.3
	0.5/0.05	3/3	3/3	3/3	-	-	-	1.9	0.0	0.4
	1/0.1	3/3	3/3	3/3	-	-	-	2.6	0.0	0.3
10/25–26	0/0	2/3	3/3	3/3	27.35	-	-	1.1	1.9	1.8
	0.5/0.05	3/3	3/3	3/3	6.29	-	-	-	1.0	1.8
	1/0.1	1/3	3/3	3/3	18.93	-	-	-	1.0	1.5
15/36–40	0/0	3/3	3/3	3/3	-	-	-	9.5	4.1	787.7
	0.5/0.05	0/3	3/3	3/3	10.64	-	-	-	3.1	250.4
	1/0.1	0/3	3/3	3/3	26.68	-	-	-	7.4	97.2
20/44–52	0/0	3/3	3/3	2/3	-	-	8.53	104.2	85.9	183.5
	0.5/0.05	1/3	3/3	2/3	23.66	-	5.06	-	16.8	200.6
	1/0.1	0/3	3/3	2/3	38.96	-	-	-	6.8	78.1
25/58–65	0/0	3/3	3/3	1/3	-	-	x	124.3	66.6	114.1
	0.5/0.05	0/3	3/3	1/3	31.12	-	x	-	87.4	158.8
	1/0.1	0/3	3/3	1/3	42.43	-	2.74	-	99.4	122.6
30/70–76	0/0	3/3	3/3	1/3	0.54	-	x	277.3	48.8	76.6
	0.5/0.05	1/3	3/3	2/3	29.59	-	x	-	40.1	315.4
	1/0.1	0/3	3/3	2/3	49.56	-	x	-	47.3	328.7
35/86–90	0/0	2/3	2/3	0/3	1.08	3.78	x	671.0	230.9	-
	0.5/0.05	1/3	2/3	1/3	35.04	1.30	x	-	93.4	951.8
	1/0.1	0/3	2/3	1/3	53.87	1.03	x	-	117.3	1317.5
40/92–101	0/0	0/3	3/3	0/3	3.10	-	x	-	295.2	-
	0.5/0.05	1/3	3/3	1/3	39.39	-	x	-	655.6	2127.5
	1/0.1	0/3	3/3	1/3	56.71	-	x	-	811.4	1624.4
Tot.		**24/72**	**69/72**	**45/72**						
Avg.					**29.72**	**2.03**	**4.76**	**109.4**	**111.9**	**269.5**

A study of the linear relaxation of the formulations confirms the performance of the approaches. In Table 3, we examine the quality of the lower bound of the formulation at the root node. For each method, we report the gap at the root node calculated w.r.t. the best-known upper bound over all the approaches.

As expected, the vehicle-flow formulation has the weakest relaxation. Surprisingly, however, the set partitioning formulation used in the BCP has a weaker relaxation than the temperature-flow formulation. In Table 4 we can observe that this is mainly due to the commodity-flow components of our formulation and—to a certain extent—the valid inequalities (4), and (5). The table compares the performance of different configurations of the temperature-flow formulation, including or excluding different sets of valid inequalities, and the commodity-flow constraints and variables. It is worth nothing that an alternative, non-compact, temperature-flow formulation could be defined in the space of the $\{x_{ij}^t\}_{t\in\mathcal{T},(i,j)\in\mathcal{A}_t}$ variables, thus neglecting commodity-flow constraints and variables, if (6) are included in the formulation (see e.g., Letchford and Salazar-González (2006) for a study on the strength of these formulations for the capacitated VRP).

The results in Table 4 also show that the commodity-constraints component plays a crucial role in the formulation. The configuration excluding this part (C2 and C6) perform significantly worse than all the other configurations, which include the commodity-flow constraints and variables. Interestingly, configurations in which only a parte of the valid inequalities are included (i.e., C3 and C4) seems to perform slightly worse than the configuration without valid inequalities (C5). However, when also the aggregated rounded capacity cuts are included (C1) the formulation seems to benefit from the valid inequalities, by solving all the instances in smaller amount of computation time.

6.3. Performance comparison single-commodity case

To demonstrate the impact of the multi-commodity aspect of the MC-VRP-T, this section reports the performance of the formulations in a single-commodity setting. Table 5 reports the main statistics of our formulations on the single-commodity problems.

The results highlight that the overall number of commodities each customer requests impacts the problem complexity for all formulations. For the VFF the impact is, however, limited, as it can only consistently solve instances up to 10 customers for the single-commodity setting. For the BCP and TFF the effect is noticeable, as now both methods can easily solve problems with up to 50 customers.

To further investigate the performance of our methods on large-scale instances, we tested the performance of the BCP and TFF on instances with 100 customers.

Table 3
Gap between each method (expressed in %) w.r.t. the best known upper bound over all the methods and the root node.

$\|\mathcal{V}'\|/\|\mathcal{B}\|$	γ/δ	VFF	TFF	BCP
5/12–15	0/0	5.53	0.00	0.00
	0.5/0.05	34.69	0.00	0.00
	1/0.1	48.09	0.00	0.00
10/25–26	0/0	16.51	0.35	0.78
	0.5/0.05	52.93	0.00	0.00
	1/0.1	65.25	0.00	0.00
15/36–40	0/0	22.72	0.69	5.04
	0.5/0.05	58.50	0.19	1.62
	1/0.1	70.07	1.20	0.36
20/44–52	0/0	39.55	2.85	5.73
	0.5/0.05	66.49	1.32	3.25
	1/0.1	76.62	0.59	1.48
25/58–65	0/0	42.28	1.69	5.46
	0.5/0.05	71.75	1.51	3.94
	1/0.1	80.97	2.11	2.80
30/70–76	0/0	40.99	0.55	4.54
	0.5/0.05	73.93	0.82	2.35
	1/0.1	82.56	0.82	2.00
35/86–90	0/0	47.65	4.20	8.99
	0.5/0.05	72.99	3.52	6.03
	1/0.1	80.33	2.88	5.71
40/92–101	0/0	50.25	2.33	11.10
	0.5/0.05	75.19	1.99	5.59
	1/0.1	83.10	2.18	6.44
Avg.		**56.62**	**1.32**	**3.34**

Table 4
Performance of the temperature-flow formulation under different configurations of valid inequalities. C1: formulation (3) including all valid inequalities (i.e., (4)–(6)); C2: formulation (3) excluding the commodity-flow constraints (i.e., (3e)–(3g)) and variables $\ell_{ij}^b, b \in \mathcal{B}, (i,j) \in \mathcal{A}$, but including (4)–(6); C3: formulation (3) with (6); C4: formulation (3) with (4); C5: formulation (3); C6: formulation defined only using the $x_{ij}^t, t \in \mathcal{T}, (i,j) \in \mathcal{A}$ variables, i.e., formulation (3) excluding the commodity-flow constraints (i.e., (3e)–(3g)) and variables $\ell_{ij}^b, b \in \mathcal{B}, (i,j) \in \mathcal{A}$, but including (6). The table can be interpreted as Table 2.

		Opt (#)						Gap (%)						Time (s)					
$\|\mathcal{V}'\|/\|\mathcal{B}\|$	γ/δ	C1	C2	C3	C4	C5	C6	C1	C2	C3	C4	C5	C6	C1	C2	C3	C4	C5	C6
5/12–15	0/0	3/3	3/3	3/3	3/3	3/3	3/3	-	-	-	-	-	-	0.1	0.3	0.1	0.1	0.1	0.9
	0.5/0.05	3/3	3/3	3/3	3/3	3/3	3/3	-	-	-	-	-	-	0.1	0.4	0.0	0.1	0.0	0.7
	1/0.1	3/3	3/3	3/3	3/3	3/3	3/3	-	-	-	-	-	-	0.1	0.2	0.1	0.1	0.0	0.5
10/25–26	0/0	3/3	3/3	3/3	3/3	3/3	1/3	-	-	-	-	-	32.05	1.5	7.9	5.0	1.9	1.9	564.7
	0.5/0.05	3/3	3/3	3/3	3/3	3/3	1/3	-	-	-	-	-	20.46	1.9	6.2	7.3	1.3	1.0	44.9
	1/0.1	3/3	3/3	3/3	3/3	3/3	1/3	-	-	-	-	-	14.70	1.7	4.5	8.7	1.8	1.0	23.0
15/36–40	0/0	3/3	2/3	3/3	3/3	3/3	0/3	-	5.29	-	-	-	26.56	4.6	34.3	7.2	7.0	4.1	-
	0.5/0.05	3/3	2/3	3/3	3/3	3/3	1/3	-	2.29	-	-	-	25.08	2.8	31.0	7.8	4.0	3.1	180.1
	1/0.1	3/3	3/3	3/3	3/3	3/3	1/3	-	-	-	-	-	17.12	3.7	907.2	12.6	5.1	7.4	6.0
20/44–52	0/0	3/3	2/3	3/3	3/3	3/3	0/3	-	6.64	-	-	-	28.81	30.1	169.3	288.3	38.1	85.9	-
	0.5/0.05	3/3	2/3	3/3	3/3	3/3	0/3	-	5.47	-	-	-	21.18	12.0	113.6	345.3	22.9	16.8	-
	1/0.1	3/3	2/3	3/3	3/3	3/3	0/3	-	4.70	-	-	-	19.55	25.9	68.3	203.8	63.0	6.8	-
25/58–65	0/0	3/3	1/3	3/3	3/3	3/3	0/3	-	7.59	-	-	-	25.54	46.7	20.7	454.5	82.9	66.6	-
	0.5/0.05	3/3	2/3	3/3	3/3	3/3	1/3	-	12.59	-	-	-	x	53.3	1092.0	1377.3	68.6	87.4	2266.4
	1/0.1	3/3	2/3	1/3	3/3	3/3	1/3	-	9.36	0.80	-	-	x	97.3	1083.6	28.2	245.7	99.4	677.8
Tot.		**45/45**	**36/45**	**43/45**	**45/45**	**45/45**	**16/45**												
Avg.								-	**6.84**	**0.80**	-	-	**23.26**	**18.8**	**221.8**	**190.3**	**36.2**	**25.4**	**235.6**

Table 5
Performance of the vehicle-flow and temperature-flow formulations and the BCP algorithm on the single-commodity problems.

		Opt (#)			Gap (%)		Time (s)		
$\|\mathcal{V}'\|$	γ / δ	VFF	TFF	BCP	VFF	BCP	VFF	TFF	BCP
5	0/0	3/3	3/3	3/3	-	-	0.0	0.0	0.2
	0.5/0.05	3/3	3/3	3/3	-	-	0.1	0.0	0.0
	1/0.1	3/3	3/3	3/3	-	-	0.1	0.0	0.1
10	0/0	3/3	3/3	3/3	-	-	0.2	0.0	0.2
	0.5/0.05	3/3	3/3	3/3	-	-	40.5	0.0	0.1
	1/0.1	3/3	3/3	3/3	-	-	328.3	0.0	0.2
15	0/0	3/3	3/3	3/3	-	-	44.6	0.5	2.8
	0.5/0.05	1/3	3/3	3/3	11.34	-	568.6	0.6	0.7
	1/0.1	0/3	3/3	3/3	16.71	-	-	1.4	1.6
20	0/0	2/3	3/3	3/3	8.02	-	22.9	0.2	3.3
	0.5/0.05	0/3	3/3	3/3	8.75	-	-	0.2	1.0
	1/0.1	1/3	3/3	3/3	19.33	-	1462.2	0.2	2.3
25	0/0	2/3	3/3	3/3	1.28	-	45.1	3.3	3.1
	0.5/0.05	1/3	3/3	3/3	28.78	-	3381.5	2.2	1.1
	1/0.1	0/3	3/3	3/3	29.65	-	-	2.9	1.5
30	0/0	1/3	3/3	3/3	5.99	-	86.1	3.1	9.2
	0.5/0.05	0/3	3/3	3/3	34.34	-	-	5.0	3.3
	1/0.1	0/3	3/3	3/3	46.79	-	-	5.6	18.7
35	0/0	1/3	3/3	3/3	8.03	-	718.7	4.5	17.9
	0.5/0.05	0/3	3/3	3/3	33.16	-	-	3.6	4.0
	1/0.1	0/3	3/3	3/3	47.47	-	-	4.1	7.0
40	0/0	0/3	3/3	3/3	5.59	-	-	6.3	391.8
	0.5/0.05	0/3	3/3	3/3	39.89	-	-	3.2	63.3
	1/0.1	0/3	3/3	3/3	53.82	-	-	5.2	115.8
45	0/0	0/3	3/3	3/3	12.27	-	-	5.2	877.7
	0.5/0.05	0/3	3/3	3/3	45.55	-	-	4.3	45.4
	1/0.1	0/3	3/3	3/3	57.00	-	-	4.9	92.3
50	0/0	0/3	3/3	2/3	11.29	x	-	22.0	355.7
	0.5/0.05	0/3	3/3	3/3	45.30	-	-	12.9	443.0
	1/0.1	0/3	3/3	3/3	60.39	-	-	11.5	576.8
Tot.		**30/90**	**90/90**	**89/90**					
Avg.					**30.0**	x	**253.2**	**3.8**	**98.5**

For the BCP, we tested two additional configurations: one where the BCP is warm-started with a heuristic solution (BCP+), and one where it is warm-started with the optimal solution (BCP++). The heuristic solution is obtained by first optimizing the routing while neglecting temperature costs (using the TFF), and then, for the fixed routing, optimizing the temperature decisions. The optimal solution is obtained by solving the standard TFF formulation. The results, reported in Table 6, show that BCP can obtain high-quality lower bounds, but it cannot find feasible integer solutions within the time limit, while the TFF can easily solve these problems within minutes.

While providing a strong primal bound does help the BCP, BCP++ displays the best performance among the three configurations in terms of time and gap, the algorithm still struggles to close the optimality gap within the time limit in some cases, even when the optimal solution is used as warm-start (BCP++). Interestingly, this happens for the instances with low temperature costs.

Overall, the TFF outperforms the BCP++ in terms of both solution quality and computational time, demonstrating its effectiveness for solving large-scale instances of the MC-VRP-T. This may indicate that TFF may be more effective for this problem or on this dataset.

The results may suggest that the performance of the BCP may improve when temperature costs are higher. Another aspect to consider is that, if additional side-constraints are added to the problem, such as time windows, we expect the BCP to perform better compared to the TFF, as time windows are known to be difficult to handle for compact formulations, and the BCP can easily incorporate them in the pricing problem.

6.4. Results for the MC-VRP-TA

In this section, we examine the MC-VRP-TA. First, we study the computational complexity of the problem, and then we compare solutions of the MC-VRP-T and of the MC-VRP-TA in terms of costs and structure to assess the benefits of temperature adjustment.

Table 7 compares the performance of the temperature-flow formulation of the MC-VRP-T and the MC-VRP-TA on the multi-commodity dataset with low/medium temperature cost ($\gamma = 0.5,\ \delta = 0.05$) and high temperature cost ($\gamma = 1,\ \delta = 0.1$). The case without

Table 6
Performance of temperature-flow formulation and the BCP algorithm on the single-commodity problems with 100 customers. The BCP cannot find feasible integer solutions within the time limit. Therefore, BCP gaps are calculated with respect to the optimal solution provided by the temperature-flow formulation. BCP+ and BCP++ refer to the configurations where the BCP is warm-started with a heuristic solution and an optimal solution, respectively. Their gaps are calculated with respect to the best feasible solution found by the corresponding method, which always can find one due to the warm-starting.

		Gap (%)				Time (s)			
Type	γ/δ	TFF	BCP	BCP+	BCP++	TFF	BCP	BCP+	BCP++
C	0/0	0.0	2.3	1.3	1.3	183.8	TL	TL	TL
	0.5/0.05	0.0	0.3	31.8	0.0	415.8	TL	TL	949.8
	1/0.1	0.0	0.1	1.9	0.0	162.1	TL	TL	799.0
R	0/0	0.0	2.1	1.6	1.6	227.6	TL	TL	TL
	0.5/0.05	0.0	0.8	0.0	0.0	358.4	TL	2203.7	667.3
	1/0.1	0.0	0.4	0.0	0.0	143.0	TL	3332.0	646.0
RC	0/0	0.0	2.8	2.0	2.0	162.9	TL	TL	TL
	0.5/0.05	0.0	0.4	2.25	0.0	426.9	TL	TL	743.6
	1/0.1	0.0	0.3	0.0	0.0	91.5	TL	3558.0	676.5

Table 7
Performance of the temperature-flow formulation of the MC-VRP-T and the MC-VRP-TA on the multi-commodity problems.

		Opt (#)		Gap (%)		Time (s)	
$\lvert\mathcal{V}'\rvert/\lvert\mathcal{B}\rvert$	γ/δ	MC-VRP-T	MC-VRP-TA	MC-VRP-T	MC-VRP-TA	MC-VRP-T	MC-VRP-TA
5/12–15	0.5/0.05	3/3	3/3	-	-	0.0	0.1
	1/0.1	3/3	3/3	-	-	0.0	0.1
10/25–26	0.5/0.05	3/3	3/3	-	-	1.0	23.5
	1/0.1	3/3	3/3	-	-	1.0	27.3
15/36–40	0.5/0.05	3/3	3/3	-	-	3.1	133.0
	1/0.1	3/3	3/3	-	-	7.4	410.1
20/44–52	0.5/0.05	3/3	1/3	-	2.27	16.8	750.4
	1/0.1	3/3	1/3	-	2.20	6.8	471.4
25/58–65	0.5/0.05	3/3	1/3	-	3.49	87.4	95.6
	1/0.1	3/3	1/3	-	7.63	99.4	208.4
30/70–76	0.5/0.05	3/3	0/3	-	3.17	40.1	-
	1/0.1	3/3	0/3	-	3.12	47.3	-
35/86–90	0.5/0.05	2/3	0/3	1.30	8.79	93.4	-
	1/0.1	2/3	0/3	1.03	14.45	117.3	-
40/92–101	0.5/0.05	3/3	0/3	-	14.01	655.6	-
	1/0.1	3/3	0/3	-	17.85	811.4	-
Tot.		**46/48**	**22/48**				
Avg.				**1.16**	**6.83**	**125.1**	**150.3**

temperature cost ($\gamma = 0,\ \delta = 0$) is not considered because in that case one could solve the MC-VRP-T to obtain an optimal solution of the MC-VRP-TA.

The same comparison is illustrated in Table 8, but on the single-commodity problems.

The results in Tables 7 and 8 show that the MC-VRP-TA is computationally more challenging than the MC-VRP-T for the proposed temperature-flow formulation. This confirms our analysis of the problem structure, which suggested that the MC-VRP-TA would be more complex to solve than the MC-VRP-T due to the increased interdependence between routing and temperature decisions and the larger number of variables and decision space. In particular, we cannot solve any MC-VRP-TA instance with 30 or more customers for the multi-commodity case (Table 7). This is confirmed in the single-commodity case (Table 8), where only one instance with 50 customers can be solved for the MC-VRP-TA. Overall, we can conclude that for our approach, the MC-VRP-TA is more challenging to solve than the MC-VRP-T for both the multi- and single-commodity cases.

The savings achieved by allowing temperature adjustment are illustrated in Fig. 4. It visualizes the percentage differences in total cost, routing, and temperature cost of using MC-VRP-TA instead of MC-VRP-T for the instances solved to optimality. MC-VRP-TA solutions can involve higher routing costs but always lead to significant savings in temperature costs, signaling that structurally

Table 8
Performance of the temperature-flow formulation of the MC-VRP-T and the MC-VRP-TA on the single-commodity problems.

		Opt (#)		Gap (%)	Time (s)	
$\|\mathcal{V}'\|$	γ/δ	MC-VRP-T	MC-VRP-TA	MC-VRP-TA	MC-VRP-T	MC-VRP-TA
5	0.5/0.05	3/3	3/3	-	0.0	0.0
	1/0.1	3/3	3/3	-	0.0	0.0
10	0.5/0.05	3/3	3/3	-	0.0	0.0
	1/0.1	3/3	3/3	-	0.0	0.1
15	0.5/0.05	3/3	3/3	-	0.6	9.9
	1/0.1	3/3	3/3	-	1.4	11.6
20	0.5/0.05	3/3	3/3	-	0.2	0.2
	1/0.1	3/3	3/3	-	0.2	0.4
25	0.5/0.05	3/3	3/3	-	2.2	109.8
	1/0.1	3/3	3/3	-	2.9	109.1
30	0.5/0.05	3/3	2/3	7.0	5.0	17.0
	1/0.1	3/3	2/3	7.4	5.6	18.7
35	0.5/0.05	3/3	3/3	-	3.6	254.7
	1/0.1	3/3	3/3	-	4.1	25.4
40	0.5/0.05	3/3	2/3	3.1	3.2	6.4
	1/0.1	3/3	2/3	2.6	5.2	14.6
45	0.5/0.05	3/3	2/3	1.3	4.3	271.0
	1/0.1	3/3	2/3	1.6	4.9	700.7
50	0.5/0.05	3/3	0/3	2.0	12.9	-
	1/0.1	3/3	1/3	1.9	11.5	2759.5
Tot.		**60/60**	**49/60**			
Avg.				**3.0**	**3.4**	**130.2**

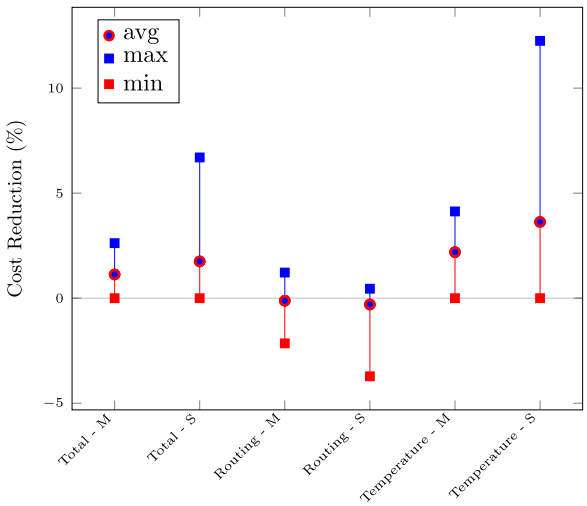
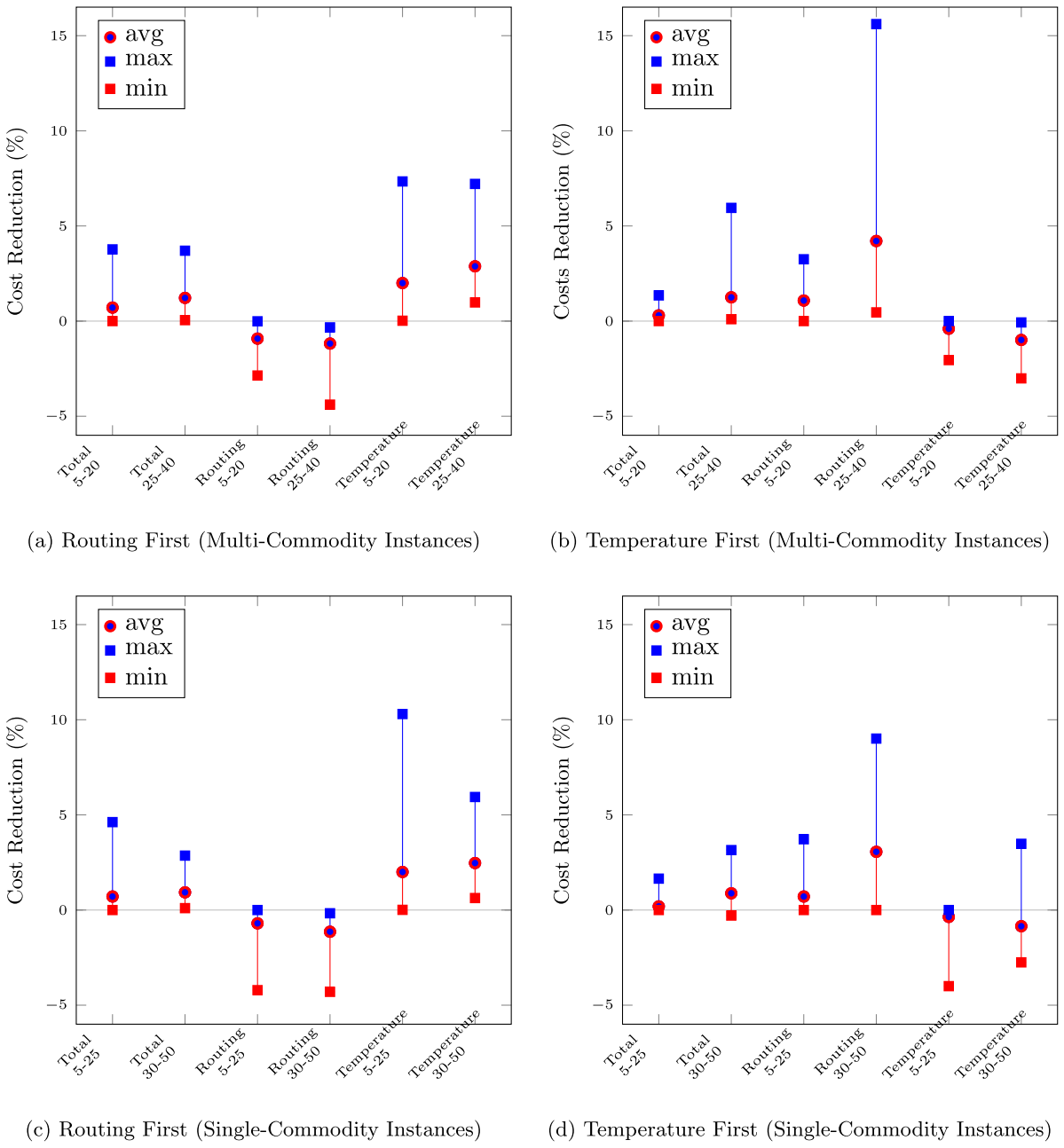


Fig. 4. Cost reductions due to adjustable temperatures (MC-VRP-T vs MC-VRP-TA). “M" and “S" denote multi-commodity and single-commodity instances, respectively.

different routes are obtained by both formulations. Additionally, we can observe that the gaps follow the same trend in both the multi- and single-commodity cases, but they are more pronounced in the latter.

6.5. Assessing the benefits of integrating routing and temperature control decisions

In this section, we demonstrate how our model can be utilized to minimize both routing and temperature costs in an integrated manner and evaluate its benefits compared to traditional sequential approaches. In particular, we solve the MC-VRP-T and compare its results to those obtained by sequentially minimizing routing costs and temperature costs (and vice versa). Firstly, feasible routes optimal w.r.t. the routing costs (temperature costs) are obtained by solving a MC-VRP-T without considering temperature-dependent



(a) Routing First (Multi-Commodity Instances)

(b) Temperature First (Multi-Commodity Instances)

(c) Routing First (Single-Commodity Instances)

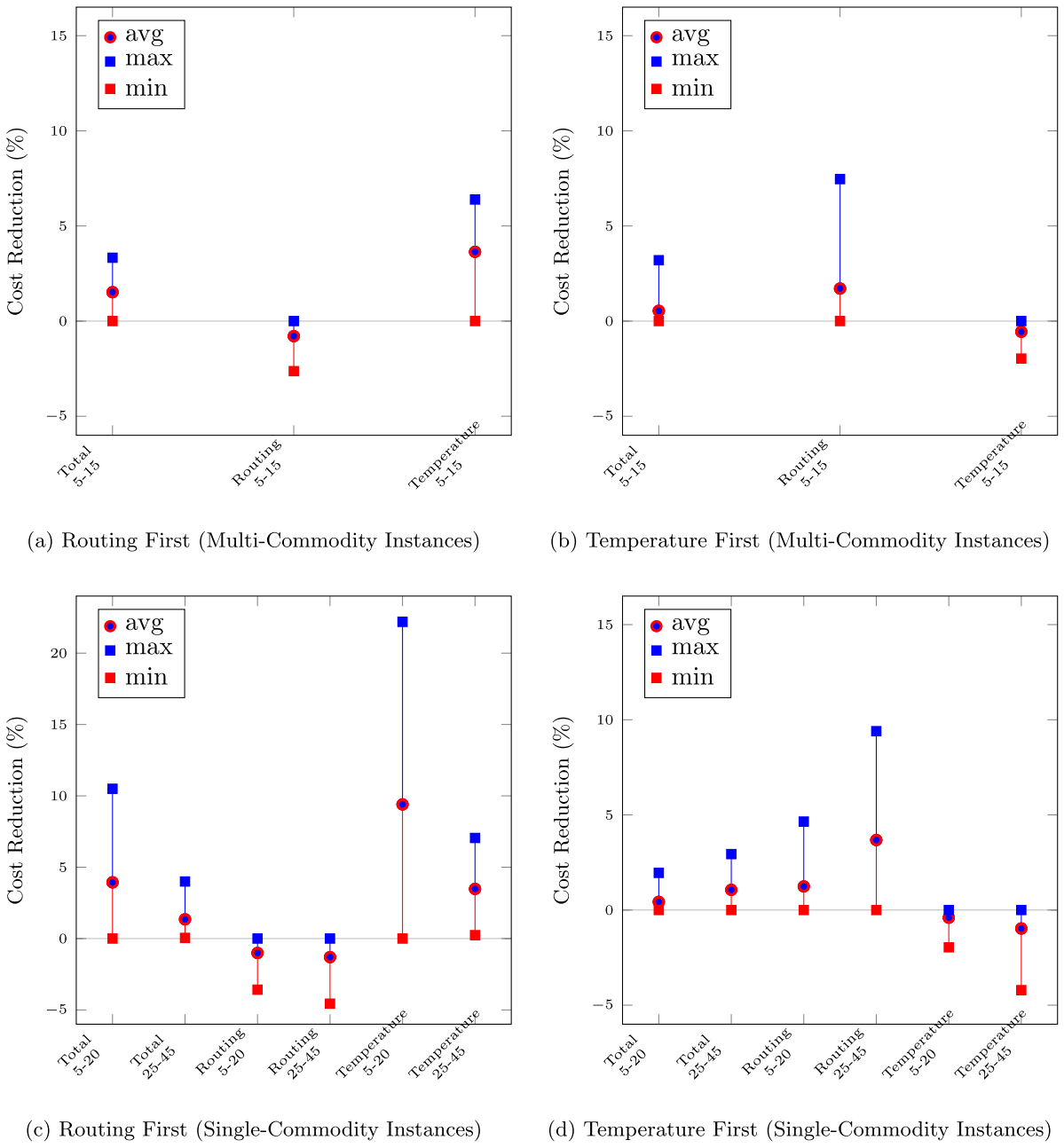
(d) Temperature First (Single-Commodity Instances)

Fig. 5. Cost reductions due to integrated decisions for the MC-VRP-T.

(routing) costs. Secondly, for the obtained routes, the MC-VRP-T is used to minimize the associated temperature (routing) costs. The same analysis is performed for the MC-VRP-TA.

Fig. 5 reports the average, maximum, and minimum percentage reduction in total cost, routing, and temperature cost by integrating routing and temperature control decisions for the instances solved to optimality. These values are calculated as the difference in the objective function value between the non-integrated and integrated models, divided by the objective function value of the integrated model.

The upper-left diagram (5a) indicates that the total cost reduction on the instances ranges between 0% and 3.8% and that average total savings tend to be larger for larger instances. For the routing cost, only cost increases are observed, which is to be expected given that the non-integrated approach first optimizes routing costs before determining the best temperature decisions. For the same reason, temperature cost savings are significantly larger in the integrated solutions. In the upper-right diagram (5b), we observe that



(a) Routing First (Multi-Commodity Instances)

(b) Temperature First (Multi-Commodity Instances)

(c) Routing First (Single-Commodity Instances)

(d) Temperature First (Single-Commodity Instances)

Fig. 6. Cost reductions due to integrated decisions for the MC-VRP-TA.

the integrated approach outperforms the sequential temperature-first approach by generating solutions with lower routing costs and a modest increase in temperature costs. Similar trends are observed for the single-commodity case, see panels (5c) and (5d).

Fig. 6 reports the same analysis for the MC-VRP-TA. Instances are clustered differently due to the difference in the instance sizes and the set of instances solved to optimality.

The results in Fig. 6 display similar trends to the fixed-temperature case (Fig. 5). However, the cost reductions seem, overall, slightly more modest in the adjustable case, which can be attributed to the increased flexibility in temperature decisions that allows for decreasing temperature costs, even when the routing is prefixed. An exception is detected in Fig. 6c), where the maximum temperature cost reduction seems quite higher for instances 5–25 average routing cost reduction is higher than in the fixed-temperature case. This signals that the routing obtained by the MC-VRP-TA is structurally different from the one obtained by the MC-VRP-T, and that it allows for larger relative temperature cost savings.

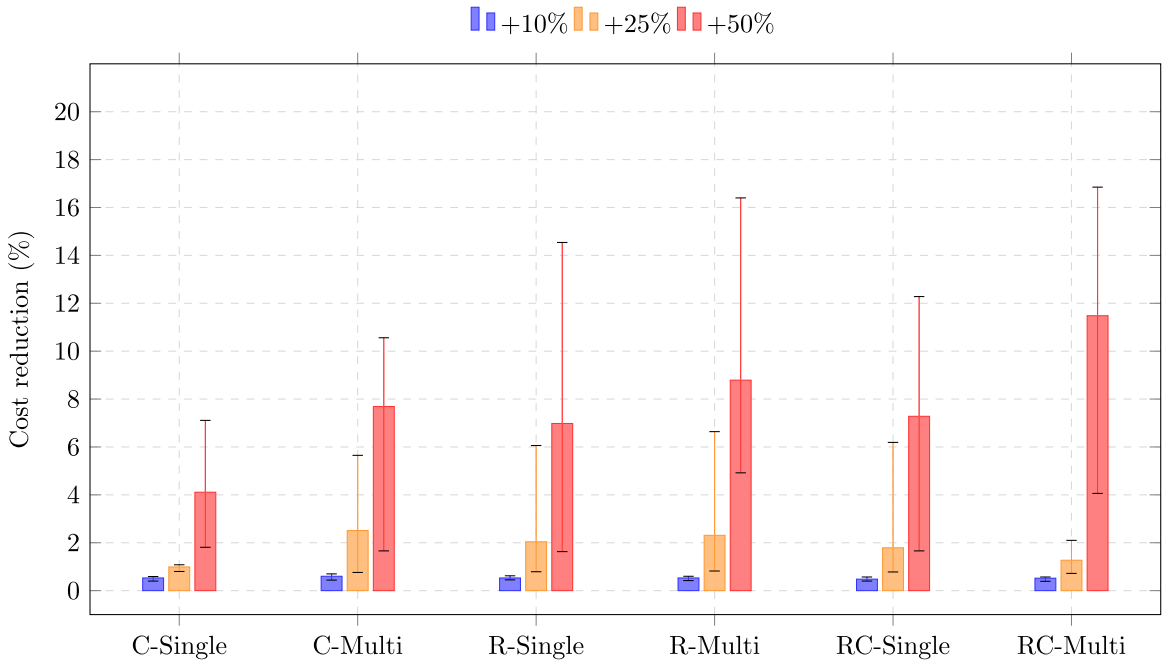


Fig. 7. Average, minimum, and maximum cost reduction for each instance class when the temperature range of every commodity is increased by 10%, 25%, and 50% w.r.t. the original instances. The error bars represent the minimum and maximum cost reduction across all instances solved to optimality in the respective class.

6.6. *Temperature range sensitivity analysis*

To provide managerial insights into how flexibility in product procurement, such as handling products with wider temperature tolerances, could save logistics costs, we perform a sensitivity analysis on the temperature ranges.

Starting from the original instances, we generate additional problems by uniformly expanding the temperature range of each commodity $i \in \mathcal{B}$ symmetrically by X. The new temperature range of each commodity $i \in \mathcal{B}$ is given by:

$$\left[\underline{T_i} - \frac{X}{2}(\overline{T_i} - \underline{T_i}),\ \overline{T_i} + \frac{X}{2}(\overline{T_i} - \underline{T_i})\right]$$

Positive values of X widen the temperature range (more relaxed requirements), while negative values narrow it (more restrictive requirements). All other instance attributes remain identical.

We generated new instances for X equal to -10%, -25%, -50%, $+10\%$, $+25\%$, and $+50\%$ for both the multi- and single-commodity cases. These new instances were solved for the MC-VRP-T using our BC algorithm on the temperature-flow formulation, and the total costs were compared with those of the original instances. The results for the relaxed temperature requirements are presented in Fig. 7, which illustrates the average cost reduction for each instance class when the temperature range is increased by 10%, 25%, and 50%. The error bars represent the minimum and maximum cost reductions across all instances solved to optimality in each class.

Fig. 7 shows that relaxing the temperature requirements leads to significant cost reductions, with an average reduction of 0.5% for a 10% increase in the temperature range, and up to 11.5% for a 50% increase. The cost reductions are, overall, more pronounced for the multicommodity instances. This is likely due to the fact that in multicommodity instances the same customer could require conflicting commodities. The impact of relaxing temperature requirements is therefore more significant in these instances, as it may allow visiting the same customer with a single vehicle, instead of multiple ones.

When the temperature range is narrowed by 10%, 25%, and 50%, almost all instances become infeasible. The exceptions are a few specific problems: in the case of instance C101n5c5, narrowing the temperature range by 10% and 25% resulted in cost increases of 0.52% and 1.03%, respectively. For instance R101n10c10, the cost increases were 0.47% and 0.87%, while for R101n5c14, the increases were 6.33% and 6.67%. Similarly, for RC101n5c5, the cost increases were 0.57% and 1.14% for 10% and 25% reductions, respectively.

The primary reason most instances become infeasible when temperature ranges are narrowed relates to the instance generation process. In the original instances, the maximum number of vehicles is tightly set by solving a bin packing problem that assigns commodities to bins (see Section 6.1). Narrowing the temperature ranges effectively reduces the compatibility between commodities. Consequently, more vehicles are needed to serve the same set of commodities. Because the maximum number of vehicles is fixed in these instances to isolate the effect of changing temperature ranges, this reduced compatibility naturally leads to infeasibility in most tightly constrained cases.

7. Conclusion

Motivated by real-life practices in cold-chain distribution, this paper introduces the MC-VRP-T, which incorporates decisions on route design and target temperatures for temperature-sensitive products. To the best of our knowledge, these decisions have not been previously integrated into the literature. As shown by our numerical experiments, significant improvements may be achieved by integrating these decisions for routing problems involving products with different storage and transportation temperature requirements.

By exploiting the structure of the problem, we show how to transform the problem into a Heterogeneous Fleet Vehicle Routing Problem and propose two new mathematical formulations. One based on flow variables, and that can be solved by means of a branch-and-cut algorithm including several valid inequalities. The other is a route-based formulation that can be solved using state-of-the-art branch-and-cut-and-price algorithms. To show the value of the proposed formulations and algorithms, extensive numerical experiments are performed, and the generated solutions are used to demonstrate the benefits of integrating temperature and routing decisions.

Moreover, we assess the benefits of adjusting target temperatures along the routes by introducing an extension of the MC-VRP-T with adjustable temperature (MC-VRP-TA). For this extension, we adapt our MC-VRP-T temperature-flow formulation to solve the MC-VRP-TA. Numerical results show that the ability to adjust cargo temperatures leads to a reduction of total costs. Moreover, we provide insights on the trade-offs between routing and temperature costs, and show that the extension is more challenging to solve.

Future research could evaluate the use of alternative objective functions, especially those that cannot be accommodated by our suggested solution approaches. For examples, if time-related constrained would be included in our problem (e.g., time windows, synchronization), the proposed formulations would need to be adapted to properly capture the refrigeration costs due to waiting times, which would likely require the introduction of additional variables and constraints. Finally, while our current methodology effectively leverages state-of-the-art approaches through problem-specific transformations, there is room for further algorithmic enhancements by developing more specialized methods, particularly to address the increased computational complexity posed by the MC-VRP-TA.

CRediT authorship contribution statement

Rosario Paradiso: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Said Dabia:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition; **Wout Dullaert:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization; **Joaquim Gromicho:** Writing – review & editing, Writing – original draft, Investigation, Funding acquisition, Data curation, Conceptualization.

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Data availability

Data will be made available on request.

Appendix A. Detailed computational results

In this section, we report the detailed computational results for the MC-VRP-T and for the MC-VRP-TA. Each table reports the Solomon instance type (Network), the number of customer nodes ($|\mathcal{V}'|$), the number of commodities $|\mathcal{B}|$, and the values of γ and δ. The next columns report the best feasible solution value found (BUB), the best lower bound value obtained (BLB) and the optimality gap (Gap). The last two columns report the computational time (Time (s)) and the root node lower bound (RN-LB), respectively. Tables A.1, A.2, A.3, A.4, A.5, A.6, A.7, A.8, A.9 are dedicated to the MC-VRP-T, while Tables A.10, A.11, A.12 report the results for the MC-VRP-TA.

Table A.1
Detailed computational results of the vehicle-flow formulation of the MC-VRP-T on the C instances.

Network	$\lvert\mathcal{V}'\rvert$	$\lvert\mathcal{B}\rvert$	γ	δ	BUB	BLB	Gap	Time (s)	RN-LB
C	5	11	0	0	128.00	128.00	0.00%	0.1	119.00
C	5	11	0.5	0.05	226.75	226.75	0.00%	0.5	153.46
C	5	11	1	0.1	325.50	325.50	0.00%	0.9	177.80
C	10	24	0	0	185.00	185.00	0.00%	1.6	178.00
C	10	24	0.5	0.05	382.85	382.85	0.00%	142.3	183.21
C	10	24	1	0.1	580.70	389.29	32.96%	TL	190.73
C	15	39	0	0	340.00	340.00	0.00%	8.8	258.00
C	15	39	0.5	0.05	700.40	565.35	19.28%	TL	267.25
C	15	39	1	0.1	1050.80	770.74	26.65%	TL	272.97
C	20	51	0	0	417.00	417.00	0.00%	247.5	203.00
C	20	51	0.5	0.05	868.90	612.84	29.47%	TL	238.54
C	20	51	1	0.1	1309.20	738.66	43.58%	TL	258.64
C	25	61	0	0	413.00	413.00	0.00%	319.2	207.00
C	25	61	0.5	0.05	924.55	533.57	42.29%	TL	208.83
C	25	61	1	0.1	1427.20	652.55	54.28%	TL	196.51
C	30	75	0	0	452.00	452.00	0.00%	457.0	273.00
C	30	75	0.5	0.05	1062.35	620.68	41.58%	TL	227.76
C	30	75	1	0.1	1672.60	725.30	56.64%	TL	237.98
C	35	88	0	0	651.00	643.35	1.08%	TL	303.00
C	35	88	0.5	0.05	1485.20	884.34	40.46%	TL	317.30
C	35	88	1	0.1	2285.00	1060.51	53.59%	TL	384.42
C	40	96	0	0	676.00	648.33	3.99%	TL	324.00
C	40	96	0.5	0.05	1587.80	901.51	43.22%	TL	339.11
C	40	96	1	0.1	2512.80	1070.70	57.39%	TL	340.41
C	5	5	0	0	116.00	116.00	0.00%	0.1	114.00
C	5	5	0.5	0.05	183.05	183.04	0.01%	0.3	127.80
C	5	5	1	0.1	250.10	250.10	0.01%	0.4	148.63
C	10	10	0	0	129.00	129.00	0.00%	0.2	125.00
C	10	10	0.5	0.05	247.90	247.90	0.00%	104.9	127.17
C	10	10	1	0.1	366.80	366.77	0.01%	874.3	129.38
C	15	15	0	0	274.00	274.00	0.00%	4.8	199.00
C	15	15	0.5	0.05	468.95	388.11	17.24%	TL	197.56
C	15	15	1	0.1	658.10	476.80	27.55%	TL	199.60
C	20	20	0	0	324.00	297.62	8.02%	TL	158.00
C	20	20	0.5	0.05	607.90	548.75	9.73%	TL	194.29
C	20	20	1	0.1	882.00	730.52	17.18%	TL	194.41
C	25	25	0	0	313.00	308.07	1.28%	TL	166.00
C	25	25	0.5	0.05	614.55	457.51	25.55%	TL	176.83
C	25	25	1	0.1	948.50	491.83	48.15%	TL	191.42
C	30	30	0	0	356.00	345.47	2.81%	TL	184.00
C	30	30	0.5	0.05	748.75	460.42	38.51%	TL	190.00
C	30	30	1	0.1	1124.00	525.15	53.28%	TL	199.02
C	35	35	0	0	433.00	419.02	3.00%	TL	217.00
C	35	35	0.5	0.05	875.50	580.86	33.65%	TL	220.81
C	35	35	1	0.1	1309.00	707.74	45.93%	TL	220.85
C	40	40	0	0	494.00	482.94	2.23%	TL	228.00
C	40	40	0.5	0.05	1016.90	623.28	38.71%	TL	257.66
C	40	40	1	0.1	1488.60	718.13	51.76%	TL	285.23
C	45	45	0	0	576.00	500.83	13.02%	TL	253.00
C	45	45	0.5	0.05	1230.40	652.48	46.97%	TL	277.69
C	45	45	1	0.1	1817.60	755.49	58.43%	TL	253.28
C	50	50	0	0	609.00	539.58	11.33%	TL	275.00
C	50	50	0.5	0.05	1267.60	697.03	45.01%	TL	289.52
C	50	50	1	0.1	2054.30	786.30	61.72%	TL	288.99

Table A.2
Detailed computational results of the vehicle-flow formulation of the MC-VRP-T on the R instances.

Network	$\lvert\mathcal{V}'\rvert$	$\lvert\mathcal{B}\rvert$	γ	δ	BUB	BLB	Gap	Time (s)	RN-LB
R	5	14	0	0	312.00	312.00	0.00%	0.2	312.00
R	5	14	0.5	0.05	567.50	567.50	0.00%	1.7	361.44
R	5	14	1	0.1	823.00	823.00	0.00%	2.8	416.35
R	10	25	0	0	473.00	473.00	0.00%	0.7	385.00
R	10	25	0.5	0.05	831.60	791.24	4.85%	TL	417.20
R	10	25	1	0.1	1185.20	1185.20	0.00%	95.5	448.09
R	15	35	0	0	628.00	628.00	0.00%	5.6	526.00
R	15	35	0.5	0.05	1160.00	1064.87	8.20%	TL	563.36
R	15	35	1	0.1	1681.60	1348.57	19.80%	TL	584.24
R	20	43	0	0	680.00	680.00	0.00%	6.1	543.00
R	20	43	0.5	0.05	1347.35	1181.94	12.28%	TL	573.47
R	20	43	1	0.1	2006.70	1441.31	28.18%	TL	588.04
R	25	57	0	0	789.00	789.00	0.00%	19.3	587.00
R	25	57	0.5	0.05	1582.05	1325.76	16.20%	TL	592.33
R	25	57	1	0.1	2364.50	1801.60	23.81%	TL	623.88
R	30	69	0	0	841.00	841.00	0.00%	97.5	590.00
R	30	69	0.5	0.05	1714.50	1581.78	7.74%	TL	607.45
R	30	69	1	0.1	2569.00	1542.21	39.97%	TL	591.22
R	35	85	0	0	1065.00	1065.00	0.00%	228.3	676.00
R	35	85	0.5	0.05	2196.75	1896.71	13.66%	TL	712.08
R	35	85	1	0.1	3318.20	1986.14	40.14%	TL	718.63
R	40	91	0	0	1132.00	1130.67	0.09%	TL	697.00
R	40	91	0.5	0.05	2313.65	1690.78	26.92%	TL	766.08
R	40	91	1	0.1	3430.00	1892.24	44.83%	TL	797.51
R	5	5	0	0	161.00	161.00	0.00%	0.0	161.00
R	5	5	0.5	0.05	257.75	257.75	0.00%	0.0	253.24
R	5	5	1	0.1	354.50	354.50	0.00%	0.0	345.55
R	10	10	0	0	293.00	293.00	0.00%	0.2	233.00
R	10	10	0.5	0.05	528.35	528.35	0.00%	2.4	299.37
R	10	10	1	0.1	762.70	762.70	0.00%	53.6	314.83
R	15	15	0	0	390.00	390.00	0.00%	0.3	304.00
R	15	15	0.5	0.05	732.40	732.40	0.00%	568.6	325.47
R	15	15	1	0.1	1069.80	928.40	13.22%	TL	332.93
R	20	20	0	0	483.00	483.00	0.00%	2.8	334.00
R	20	20	0.5	0.05	870.30	777.66	10.65%	TL	355.65
R	20	20	1	0.1	1248.60	1248.48	0.01%	1462.2	360.96
R	25	25	0	0	584.00	584.00	0.00%	82.1	376.00
R	25	25	0.5	0.05	1037.50	1037.40	0.01%	3381.5	394.97
R	25	25	1	0.1	1488.00	1358.34	8.71%	TL	399.75
R	30	30	0	0	624.00	624.00	0.00%	86.1	376.00
R	30	30	0.5	0.05	1162.55	874.81	24.75%	TL	405.65
R	30	30	1	0.1	1687.10	957.60	43.24%	TL	397.08
R	35	35	0	0	654.00	654.00	0.00%	718.7	420.00
R	35	35	0.5	0.05	1238.80	973.32	21.43%	TL	454.98
R	35	35	1	0.1	1820.60	1082.06	40.57%	TL	455.52
R	40	40	0	0	741.00	731.17	1.21%	TL	469.00
R	40	40	0.5	0.05	1404.00	920.72	34.42%	TL	493.03
R	40	40	1	0.1	2063.00	1060.58	48.59%	TL	494.56
R	45	45	0	0	835.00	793.18	4.91%	TL	501.00
R	45	45	0.5	0.05	1603.80	1062.46	33.75%	TL	520.73
R	45	45	1	0.1	2332.60	1152.85	50.58%	TL	524.52
R	50	50	0	0	852.00	813.58	4.46%	TL	514.00
R	50	50	0.5	0.05	1708.15	1113.97	34.79%	TL	524.79
R	50	50	1	0.1	2572.80	1249.58	51.43%	TL	529.53

Table A.3
Detailed computational results of the vehicle-flow formulation of the MC-VRP-T on the RC instances.

Network	$\lvert\mathcal{V}'\rvert$	$\lvert\mathcal{B}\rvert$	γ	δ	BUB	BLB	Gap	Time (s)	RN-LB
RC	5	12	0	0	324.00	324.00	0.00%	0.2	293.00
RC	5	12	0.5	0.05	485.15	485.15	0.00%	3.6	313.17
RC	5	12	1	0.1	646.30	646.30	0.00%	4.1	326.60
RC	10	24	0	0	501.00	363.85	27.35%	TL	341.00
RC	10	24	0.5	0.05	817.50	703.10	13.99%	TL	352.94
RC	10	24	1	0.1	1164.40	886.93	23.83%	TL	391.28
RC	15	38	0	0	529.00	529.00	0.00%	14.2	382.00
RC	15	38	0.5	0.05	978.35	934.86	4.45%	TL	369.52
RC	15	38	1	0.1	1425.80	947.14	33.57%	TL	414.46
RC	20	51	0	0	678.00	678.00	0.00%	59.0	358.00
RC	20	51	0.5	0.05	1252.20	885.98	29.25%	TL	382.00
RC	20	51	1	0.1	1822.40	999.89	45.13%	TL	383.85
RC	25	64	0	0	777.00	777.00	0.00%	34.4	378.00
RC	25	64	0.5	0.05	1558.00	1014.93	34.86%	TL	384.95
RC	25	64	1	0.1	2334.50	1185.71	49.21%	TL	395.51
RC	30	73	0	0	1117.00	1110.17	0.54%	TL	519.00
RC	30	73	0.5	0.05	2065.70	1250.99	39.44%	TL	440.78
RC	30	73	1	0.1	3004.40	1440.02	52.07%	TL	450.82
RC	35	89	0	0	1191.00	1191.00	0.00%	1113.7	560.00
RC	35	89	0.5	0.05	2368.55	1160.71	51.00%	TL	639.42
RC	35	89	1	0.1	3539.10	1136.86	67.88%	TL	723.37
RC	40	100	0	0	1167.00	1105.08	5.23%	TL	464.00
RC	40	100	0.5	0.05	2331.40	1211.46	48.04%	TL	455.32
RC	40	100	1	0.1	3787.10	1215.95	67.89%	TL	479.87
RC	5	5	0	0	169.00	169.00	0.00%	0.0	164.00
RC	5	5	0.5	0.05	244.95	244.95	0.00%	0.1	234.34
RC	5	5	1	0.1	320.90	320.90	0.00%	0.1	289.02
RC	10	10	0	0	284.00	284.00	0.00%	0.3	240.00
RC	10	10	0.5	0.05	435.30	435.29	0.00%	14.1	260.51
RC	10	10	1	0.1	584.60	584.60	0.00%	56.8	294.71
RC	15	15	0	0	463.00	463.00	0.00%	128.6	316.00
RC	15	15	0.5	0.05	742.80	702.42	5.44%	TL	322.10
RC	15	15	1	0.1	1021.60	925.85	9.37%	TL	321.76
RC	20	20	0	0	540.00	540.00	0.00%	43.0	360.00
RC	20	20	0.5	0.05	948.05	892.42	5.87%	TL	369.57
RC	20	20	1	0.1	1356.10	1064.84	21.48%	TL	344.25
RC	25	25	0	0	565.00	565.00	0.00%	8.1	303.00
RC	25	25	0.5	0.05	1028.60	699.37	32.01%	TL	306.05
RC	25	25	1	0.1	1490.40	1012.08	32.09%	TL	317.28
RC	30	30	0	0	839.00	761.47	9.18%	TL	393.00
RC	30	30	0.5	0.05	1497.85	902.11	39.77%	TL	398.47
RC	30	30	1	0.1	2147.70	1206.04	43.85%	TL	405.14
RC	35	35	0	0	934.00	811.44	13.06%	TL	421.00
RC	35	35	0.5	0.05	1726.45	960.12	44.39%	TL	432.46
RC	35	35	1	0.1	2528.80	1114.75	55.92%	TL	434.59
RC	40	40	0	0	931.00	806.07	13.32%	TL	411.00
RC	40	40	0.5	0.05	1680.30	898.45	46.53%	TL	426.42
RC	40	40	1	0.1	2514.20	977.96	61.10%	TL	426.84
RC	45	45	0	0	1102.00	893.64	18.87%	TL	461.00
RC	45	45	0.5	0.05	2024.45	892.18	55.93%	TL	436.30
RC	45	45	1	0.1	2890.40	1098.99	61.98%	TL	436.70
RC	50	50	0	0	1062.00	869.40	18.08%	TL	448.00
RC	50	50	0.5	0.05	2022.80	888.00	56.10%	TL	453.00
RC	50	50	1	0.1	3227.10	1032.16	68.02%	TL	447.26

Table A.4
Detailed computational results of the temperature-flow formulation of the MC-VRP-T on the C instances.

Network	$\lvert\mathcal{V}'\rvert$	$\lvert\mathcal{B}\rvert$	γ	δ	BUB	BLB	Gap	Time (s)	RN-LB
C	5	11	0	0	128.00	128.00	0.00%	0.0	128.00
C	5	11	0.5	0.05	226.75	226.75	0.00%	0.0	226.75
C	5	11	1	0.1	325.50	325.50	0.00%	0.0	325.50
C	10	24	0	0	185.00	185.00	0.00%	0.3	185.00
C	10	24	0.5	0.05	382.85	382.85	0.00%	0.4	382.85
C	10	24	1	0.1	580.70	580.70	0.00%	0.4	580.70
C	15	39	0	0	340.00	340.00	0.00%	2.0	340.00
C	15	39	0.5	0.05	700.40	700.40	0.00%	2.6	700.40
C	15	39	1	0.1	1050.80	1050.80	0.00%	3.0	1049.50
C	20	51	0	0	417.00	417.00	0.00%	222.7	401.00
C	20	51	0.5	0.05	868.90	868.90	0.00%	36.5	851.55
C	20	51	1	0.1	1309.20	1309.20	0.00%	8.5	1300.80
C	25	61	0	0	413.00	413.00	0.00%	119.6	408.00
C	25	61	0.5	0.05	924.55	924.55	0.00%	136.4	902.15
C	25	61	1	0.1	1427.20	1427.20	0.00%	116.9	1381.50
C	30	75	0	0	452.00	452.00	0.00%	47.0	449.00
C	30	75	0.5	0.05	1062.35	1062.35	0.00%	44.6	1055.80
C	30	75	1	0.1	1662.70	1662.70	0.00%	42.2	1650.10
C	35	88	0	0	651.00	651.00	0.00%	368.2	621.00
C	35	88	0.5	0.05	1468.70	1468.70	0.00%	155.4	1424.95
C	35	88	1	0.1	2274.20	2274.20	0.00%	186.5	2234.60
C	40	96	0	0	676.00	676.00	0.00%	596.3	659.00
C	40	96	0.5	0.05	1583.95	1583.95	0.00%	1552.6	1535.25
C	40	96	1	0.1	2482.90	2482.90	0.00%	1289.2	2403.90
C	5	5	0	0	116.00	116.00	0.00%	0.0	116.00
C	5	5	0.5	0.05	183.05	183.05	0.00%	0.0	183.05
C	5	5	1	0.1	250.10	250.10	0.00%	0.0	250.10
C	10	10	0	0	129.00	129.00	0.00%	0.0	129.00
C	10	10	0.5	0.05	247.90	247.90	0.00%	0.0	247.90
C	10	10	1	0.1	366.80	366.80	0.00%	0.0	366.80
C	15	15	0	0	274.00	274.00	0.00%	0.0	274.00
C	15	15	0.5	0.05	468.95	468.95	0.00%	0.1	468.95
C	15	15	1	0.1	658.10	658.10	0.00%	0.1	658.10
C	20	20	0	0	323.00	323.00	0.00%	0.1	323.00
C	20	20	0.5	0.05	607.90	607.90	0.00%	0.1	607.90
C	20	20	1	0.1	882.00	882.00	0.00%	0.1	882.00
C	25	25	0	0	313.00	313.00	0.00%	6.2	307.00
C	25	25	0.5	0.05	614.55	614.55	0.00%	2.6	601.35
C	25	25	1	0.1	914.20	914.20	0.00%	5.2	900.60
C	30	30	0	0	356.00	356.00	0.00%	2.1	356.00
C	30	30	0.5	0.05	745.45	745.45	0.00%	3.5	745.45
C	30	30	1	0.1	1124.00	1124.00	0.00%	4.3	1123.90
C	35	35	0	0	433.00	433.00	0.00%	1.3	433.00
C	35	35	0.5	0.05	875.50	875.50	0.00%	1.9	875.50
C	35	35	1	0.1	1309.00	1309.00	0.00%	1.6	1309.00
C	40	40	0	0	494.00	494.00	0.00%	5.0	494.00
C	40	40	0.5	0.05	992.50	992.50	0.00%	2.2	992.50
C	40	40	1	0.1	1488.60	1488.60	0.00%	4.9	1488.60
C	45	45	0	0	576.00	576.00	0.00%	5.5	574.00
C	45	45	0.5	0.05	1201.20	1201.20	0.00%	5.5	1193.95
C	45	45	1	0.1	1817.60	1817.60	0.00%	7.6	1807.10
C	50	50	0	0	606.00	606.00	0.00%	30.2	587.00
C	50	50	0.5	0.05	1266.90	1266.90	0.00%	17.8	1245.85
C	50	50	1	0.1	1914.00	1914.00	0.00%	11.8	1890.10

Table A.5
Detailed computational results of the temperature-flow formulation of the MC-VRP-T on the R instances.

Network	$\lvert\mathcal{V}'\rvert$	$\lvert\mathcal{B}\rvert$	γ	δ	BUB	BLB	Gap	Time (s)	RN-LB
R	5	14	0	0	312.00	312.00	0.00%	0.1	312.00
R	5	14	0.5	0.05	567.50	567.50	0.00%	0.1	567.50
R	5	14	1	0.1	823.00	823.00	0.00%	0.1	823.00
R	10	25	0	0	473.00	473.00	0.00%	5.1	468.00
R	10	25	0.5	0.05	831.60	831.60	0.00%	2.1	831.60
R	10	25	1	0.1	1185.20	1185.20	0.00%	1.8	1185.20
R	15	35	0	0	628.00	628.00	0.00%	0.7	628.00
R	15	35	0.5	0.05	1160.00	1160.00	0.00%	0.6	1160.00
R	15	35	1	0.1	1681.60	1681.60	0.00%	0.5	1681.60
R	20	43	0	0	680.00	680.00	0.00%	6.3	666.00
R	20	43	0.5	0.05	1347.35	1347.35	0.00%	5.0	1331.10
R	20	43	1	0.1	2006.70	2006.70	0.00%	5.4	1983.90
R	25	57	0	0	789.00	789.00	0.00%	13.4	787.00
R	25	57	0.5	0.05	1582.05	1582.05	0.00%	11.4	1570.75
R	25	57	1	0.1	2364.50	2364.50	0.00%	11.3	2347.30
R	30	69	0	0	841.00	841.00	0.00%	14.4	841.00
R	30	69	0.5	0.05	1714.50	1714.50	0.00%	24.9	1702.90
R	30	69	1	0.1	2569.00	2569.00	0.00%	24.4	2569.00
R	35	85	0	0	1065.00	1065.00	0.00%	93.7	1055.00
R	35	85	0.5	0.05	2196.75	2196.75	0.00%	31.5	2182.15
R	35	85	1	0.1	3318.20	3318.20	0.00%	48.0	3293.10
R	40	91	0	0	1132.00	1132.00	0.00%	62.0	1127.00
R	40	91	0.5	0.05	2287.55	2287.55	0.00%	48.8	2285.70
R	40	91	1	0.1	3430.00	3430.00	0.00%	68.8	3409.60
R	5	5	0	0	161.00	161.00	0.00%	0.0	161.00
R	5	5	0.5	0.05	257.75	257.75	0.00%	0.0	257.75
R	5	5	1	0.1	354.50	354.50	0.00%	0.0	354.50
R	10	10	0	0	293.00	293.00	0.00%	0.0	293.00
R	10	10	0.5	0.05	528.35	528.35	0.00%	0.0	528.35
R	10	10	1	0.1	762.70	762.70	0.00%	0.0	762.70
R	15	15	0	0	390.00	390.00	0.00%	0.1	390.00
R	15	15	0.5	0.05	732.40	732.40	0.00%	0.1	732.40
R	15	15	1	0.1	1069.80	1069.80	0.00%	0.1	1069.80
R	20	20	0	0	483.00	483.00	0.00%	0.2	483.00
R	20	20	0.5	0.05	870.30	870.30	0.00%	0.3	870.30
R	20	20	1	0.1	1248.60	1248.60	0.00%	0.2	1248.60
R	25	25	0	0	584.00	584.00	0.00%	3.4	584.00
R	25	25	0.5	0.05	1037.50	1037.50	0.00%	3.1	1037.50
R	25	25	1	0.1	1488.00	1488.00	0.00%	2.4	1488.00
R	30	30	0	0	624.00	624.00	0.00%	1.0	624.00
R	30	30	0.5	0.05	1162.55	1162.55	0.00%	2.4	1162.55
R	30	30	1	0.1	1687.10	1687.10	0.00%	3.9	1687.10
R	35	35	0	0	654.00	654.00	0.00%	2.5	654.00
R	35	35	0.5	0.05	1238.80	1238.80	0.00%	1.3	1238.80
R	35	35	1	0.1	1820.60	1820.60	0.00%	1.3	1820.60
R	40	40	0	0	741.00	741.00	0.00%	11.6	741.00
R	40	40	0.5	0.05	1404.00	1404.00	0.00%	5.6	1404.00
R	40	40	1	0.1	2063.00	2063.00	0.00%	8.6	2063.00
R	45	45	0	0	835.00	835.00	0.00%	8.4	835.00
R	45	45	0.5	0.05	1584.80	1584.80	0.00%	4.0	1584.80
R	45	45	1	0.1	2332.60	2332.60	0.00%	4.9	2332.60
R	50	50	0	0	852.00	852.00	0.00%	25.9	852.00
R	50	50	0.5	0.05	1708.15	1708.15	0.00%	9.0	1708.15
R	50	50	1	0.1	2552.30	2552.30	0.00%	13.9	2552.30

Table A.6
Detailed computational results of the temperature-flow formulation of the MC-VRP-T on the RC instances.

Network	$\lvert\mathcal{V}'\rvert$	$\lvert\mathcal{B}\rvert$	γ	δ	BUB	BLB	Gap	Time (s)	RN-LB
RC	5	12	0	0	324.00	324.00	0.00%	0.0	324.00
RC	5	12	0.5	0.05	485.15	485.15	0.00%	0.0	485.15
RC	5	12	1	0.1	646.30	646.30	0.00%	0.0	646.30
RC	10	24	0	0	468.00	468.00	0.00%	0.3	468.00
RC	10	24	0.5	0.05	817.50	817.50	0.00%	0.6	817.50
RC	10	24	1	0.1	1164.40	1164.40	0.00%	0.8	1164.40
RC	15	38	0	0	529.00	529.00	0.00%	9.5	518.00
RC	15	38	0.5	0.05	978.35	978.35	0.00%	6.2	972.80
RC	15	38	1	0.1	1425.80	1425.80	0.00%	18.7	1376.30
RC	20	51	0	0	678.00	678.00	0.00%	28.6	660.00
RC	20	51	0.5	0.05	1252.20	1252.20	0.00%	9.0	1242.60
RC	20	51	1	0.1	1819.60	1819.60	0.00%	6.4	1819.60
RC	25	64	0	0	777.00	777.00	0.00%	66.8	749.00
RC	25	64	0.5	0.05	1558.00	1558.00	0.00%	114.4	1536.30
RC	25	64	1	0.1	2334.50	2334.50	0.00%	170.1	2278.40
RC	30	73	0	0	1117.00	1117.00	0.00%	84.9	1106.00
RC	30	73	0.5	0.05	2065.70	2065.70	0.00%	50.7	2041.55
RC	30	73	1	0.1	3004.40	3004.40	0.00%	75.3	2953.10
RC	35	89	0	0	1191.00	1146.00	3.78%	TL	1107.00
RC	35	89	0.5	0.05	2368.55	2337.80	1.30%	TL	2204.85
RC	35	89	1	0.1	3539.10	3502.80	1.03%	TL	3321.50
RC	40	100	0	0	1167.00	1167.00	0.00%	227.3	1120.00
RC	40	100	0.5	0.05	2331.40	2331.15	0.01%	365.4	2265.50
RC	40	100	1	0.1	3491.80	3491.55	0.01%	1076.3	3395.50
RC	5	5	0	0	169.00	169.00	0.00%	0.0	169.00
RC	5	5	0.5	0.05	244.95	244.95	0.00%	0.0	244.95
RC	5	5	1	0.1	320.90	320.90	0.00%	0.0	320.90
RC	10	10	0	0	284.00	284.00	0.00%	0.0	284.00
RC	10	10	0.5	0.05	435.30	435.30	0.00%	0.0	435.30
RC	10	10	1	0.1	584.60	584.60	0.00%	0.0	584.60
RC	15	15	0	0	463.00	463.00	0.00%	1.4	433.00
RC	15	15	0.5	0.05	742.80	742.80	0.00%	1.6	712.50
RC	15	15	1	0.1	1021.60	1021.60	0.00%	4.0	985.10
RC	20	20	0	0	540.00	540.00	0.00%	0.2	540.00
RC	20	20	0.5	0.05	948.05	948.05	0.00%	0.3	948.05
RC	20	20	1	0.1	1356.10	1356.10	0.00%	0.2	1356.10
RC	25	25	0	0	565.00	565.00	0.00%	0.5	565.00
RC	25	25	0.5	0.05	1028.60	1028.60	0.00%	0.9	1028.60
RC	25	25	1	0.1	1490.40	1490.40	0.00%	1.0	1490.40
RC	30	30	0	0	839.00	838.58	0.00%	6.1	838.00
RC	30	30	0.5	0.05	1489.65	1489.65	0.00%	9.0	1461.95
RC	30	30	1	0.1	2129.50	2129.50	0.00%	8.6	2113.00
RC	35	35	0	0	934.00	934.00	0.00%	9.8	931.00
RC	35	35	0.5	0.05	1707.95	1707.95	0.00%	7.7	1703.80
RC	35	35	1	0.1	2475.80	2475.80	0.00%	9.5	2475.80
RC	40	40	0	0	931.00	931.00	0.00%	2.3	931.00
RC	40	40	0.5	0.05	1642.80	1642.80	0.00%	1.8	1642.80
RC	40	40	1	0.1	2352.60	2352.60	0.00%	2.1	2352.60
RC	45	45	0	0	1102.00	1101.08	0.00%	1.7	1101.08
RC	45	45	0.5	0.05	1984.75	1984.74	0.00%	3.3	1984.74
RC	45	45	1	0.1	2860.40	2860.40	0.00%	2.2	2860.40
RC	50	50	0	0	1062.00	1062.00	0.00%	9.8	1061.00
RC	50	50	0.5	0.05	2010.85	2010.85	0.00%	12.0	2010.66
RC	50	50	1	0.1	2948.10	2948.10	0.00%	8.7	2948.10

Table A.7
Detailed computational results of the BCP algorithm on the C instances of the MC-VRP-T.

Network	$\lvert\mathcal{V}'\rvert$	$\lvert\mathcal{B}\rvert$	γ	δ	BUB	BLB	Gap	Time (s)	RN-LB
C	5	11	0	0	128.00	128.00	0.00%	0.1	128.00
C	5	11	0.5	0.05	226.75	226.75	0.00%	0.1	226.75
C	5	11	1	0.1	325.50	325.50	0.00%	0.1	325.50
C	10	24	0	0	185.00	185.00	0.00%	1.5	185.00
C	10	24	0.5	0.05	382.85	382.85	0.00%	1.6	382.85
C	10	24	1	0.1	580.70	580.70	0.00%	1.6	580.70
C	15	39	0	0	340.00	340.00	0.00%	249.7	325.38
C	15	39	0.5	0.05	700.40	700.40	0.00%	116.2	686.74
C	15	39	1	0.1	1050.80	1050.80	0.00%	87.8	1039.56
C	20	51	0	0	417.00	417.00	0.00%	329.8	410.79
C	20	51	0.5	0.05	868.90	868.90	0.00%	361.3	860.21
C	20	51	1	0.1	1309.20	1309.20	0.00%	121.5	1301.41
C	25	61	0	0	-	397.31	-	TL	389.50
C	25	61	0.5	0.05	-	914.09	-	TL	902.46
C	25	61	1	0.1	1427.20	1418.95	0.58%	TL	1396.62
C	30	75	0	0	-	406.29	-	TL	396.28
C	30	75	0.5	0.05	-	997.04	-	TL	987.46
C	30	75	1	0.1	-	1599.04	-	TL	1562.72
C	35	88	0	0	-	574.22	-	TL	574.22
C	35	88	0.5	0.05	-	1397.87	-	TL	1397.87
C	35	88	1	0.1	-	2216.05	-	TL	2182.25
C	40	96	0	0	-	587.02	-	TL	586.22
C	40	96	0.5	0.05	-	1443.15	-	TL	1443.15
C	40	96	1	0.1	-	2286.35	-	TL	2286.35
C	5	5	0	0	116.00	116.00	0.00%	0.2	116.00
C	5	5	0.5	0.05	183.05	183.05	0.00%	0.0	183.05
C	5	5	1	0.1	250.10	250.10	0.00%	0.0	250.10
C	10	10	0	0	129.00	129.00	0.00%	0.1	129.00
C	10	10	0.5	0.05	247.90	247.90	0.00%	0.1	247.90
C	10	10	1	0.1	366.80	366.80	0.00%	0.1	366.80
C	15	15	0	0	274.00	274.00	0.00%	1.1	274.00
C	15	15	0.5	0.05	468.95	468.95	0.00%	0.6	468.95
C	15	15	1	0.1	658.10	658.10	0.00%	0.6	658.10
C	20	20	0	0	323.00	323.00	0.00%	1.3	323.00
C	20	20	0.5	0.05	607.90	607.90	0.00%	1.0	607.90
C	20	20	1	0.1	882.00	882.00	0.00%	1.0	882.00
C	25	25	0	0	313.00	313.00	0.00%	1.2	313.00
C	25	25	0.5	0.05	614.55	614.55	0.00%	1.2	614.55
C	25	25	1	0.1	914.20	914.20	0.00%	1.3	914.20
C	30	30	0	0	356.00	356.00	0.00%	1.9	356.00
C	30	30	0.5	0.05	745.45	745.45	0.00%	2.0	745.45
C	30	30	1	0.1	1124.00	1124.00	0.00%	2.4	1124.00
C	35	35	0	0	433.00	433.00	0.00%	27.9	433.00
C	35	35	0.5	0.05	875.50	875.50	0.00%	4.0	875.50
C	35	35	1	0.1	1309.00	1309.00	0.00%	3.7	1309.00
C	40	40	0	0	494.00	494.00	0.00%	96.5	494.00
C	40	40	0.5	0.05	992.50	992.50	0.00%	12.7	992.50
C	40	40	1	0.1	1488.60	1488.60	0.00%	14.1	1488.60
C	45	45	0	0	576.00	576.00	0.00%	709.0	570.18
C	45	45	0.5	0.05	1201.20	1201.20	0.00%	39.7	1163.93
C	45	45	1	0.1	1817.60	1817.60	0.00%	26.1	1817.60
C	50	50	0	0	606.00	606.00	0.00%	322.8	590.23
C	50	50	0.5	0.05	1266.90	1266.90	0.00%	59.2	1243.28
C	50	50	1	0.1	1914.00	1914.00	0.00%	64.9	1882.32

Table A.8
Detailed computational results of the BCP algorithm on the R instances of the MC-VRP-T.

Network	$\lvert\mathcal{V}'\rvert$	$\lvert\mathcal{B}\rvert$	γ	δ	BUB	BLB	Gap	Time (s)	RN-LB
R	5	14	0	0	312.00	312.00	0.00%	0.1	312.00
R	5	14	0.5	0.05	567.50	567.50	0.00%	0.2	567.50
R	5	14	1	0.1	823.00	823.00	0.00%	0.2	823.00
R	10	25	0	0	473.00	473.00	0.00%	3.0	462.00
R	10	25	0.5	0.05	831.60	831.60	0.00%	2.9	831.60
R	10	25	1	0.1	1185.20	1185.20	0.00%	1.9	1185.20
R	15	35	0	0	628.00	628.00	0.00%	4.7	628.00
R	15	35	0.5	0.05	1160.00	1160.00	0.00%	6.6	1160.00
R	15	35	1	0.1	1681.60	1681.60	0.00%	4.2	1681.60
R	20	43	0	0	680.00	680.00	0.00%	37.2	669.77
R	20	43	0.5	0.05	1347.35	1347.35	0.00%	39.9	1336.65
R	20	43	1	0.1	2006.70	2006.70	0.00%	34.7	1992.03
R	25	57	0	0	789.00	789.00	0.00%	114.1	789.00
R	25	57	0.5	0.05	1582.05	1582.05	0.00%	158.8	1580.35
R	25	57	1	0.1	2364.50	2364.50	0.00%	122.6	2361.71
R	30	69	0	0	841.00	841.00	0.00%	76.6	841.00
R	30	69	0.5	0.05	1714.50	1714.50	0.00%	64.3	1714.50
R	30	69	1	0.1	2569.00	2569.00	0.00%	70.9	2569.00
R	35	85	0	0	-	1052.43	-	TL	1035.32
R	35	85	0.5	0.05	2196.75	2196.75	0.00%	951.8	2148.86
R	35	85	1	0.1	3318.20	3318.20	0.00%	1317.5	3247.89
R	40	91	0	0	-	1115.12	-	TL	1115.12
R	40	91	0.5	0.05	2287.55	2287.55	0.00%	2127.5	2278.35
R	40	91	1	0.1	3430.00	3430.00	0.00%	1624.4	3419.35
R	5	5	0	0	161.00	161.00	0.00%	0.2	161.00
R	5	5	0.5	0.05	257.75	257.75	0.00%	0.0	257.75
R	5	5	1	0.1	354.50	354.50	0.00%	0.2	354.50
R	10	10	0	0	293.00	293.00	0.00%	0.3	293.00
R	10	10	0.5	0.05	528.35	528.35	0.00%	0.1	528.35
R	10	10	1	0.1	762.70	762.70	0.00%	0.1	762.70
R	15	15	0	0	390.00	390.00	0.00%	3.8	390.00
R	15	15	0.5	0.05	732.40	732.40	0.00%	0.7	732.40
R	15	15	1	0.1	1069.80	1069.80	0.00%	0.7	1069.80
R	20	20	0	0	483.00	483.00	0.00%	4.0	483.00
R	20	20	0.5	0.05	870.30	870.30	0.00%	0.9	870.30
R	20	20	1	0.1	1248.60	1248.60	0.00%	1.1	1248.60
R	25	25	0	0	584.00	584.00	0.00%	6.0	584.00
R	25	25	0.5	0.05	1037.50	1037.50	0.00%	1.4	1037.50
R	25	25	1	0.1	1488.00	1488.00	0.00%	1.4	1488.00
R	30	30	0	0	624.00	624.00	0.00%	7.6	624.00
R	30	30	0.5	0.05	1162.55	1162.55	0.00%	1.7	1162.55
R	30	30	1	0.1	1687.10	1687.10	0.00%	1.8	1687.10
R	35	35	0	0	654.00	654.00	0.00%	13.7	654.00
R	35	35	0.5	0.05	1238.80	1238.80	0.00%	4.2	1238.80
R	35	35	1	0.1	1820.60	1820.60	0.00%	4.9	1820.56
R	40	40	0	0	741.00	741.00	0.00%	1037.0	731.73
R	40	40	0.5	0.05	1404.00	1404.00	0.00%	164.7	1393.81
R	40	40	1	0.1	2063.00	2063.00	0.00%	251.6	2036.41
R	45	45	0	0	835.00	835.00	0.00%	322.3	835.00
R	45	45	0.5	0.05	1584.80	1584.80	0.00%	41.5	1584.80
R	45	45	1	0.1	2332.60	2332.60	0.00%	45.6	2332.60
R	50	50	0	0	-	844.37	-	3611.3	835.85
R	50	50	0.5	0.05	1708.15	1708.15	0.00%	1209.6	1696.00
R	50	50	1	0.1	2552.30	2552.30	0.00%	1470.3	2519.39

Table A.9
Detailed computational results of the BCP algorithm on the RC instances of the MC-VRP-T.

Network	$\lvert\mathcal{V}'\rvert$	$\lvert\mathcal{B}\rvert$	γ	δ	BUB	BLB	Gap	Time (s)	RN-LB
RC	5	12	0	0	324.00	324.00	0.00%	0.6	324.00
RC	5	12	0.5	0.05	485.15	485.15	0.00%	0.9	485.15
RC	5	12	1	0.1	646.30	646.30	0.00%	0.7	646.30
RC	10	24	0	0	468.00	468.00	0.00%	1.0	468.00
RC	10	24	0.5	0.05	817.50	817.50	0.00%	0.9	817.50
RC	10	24	1	0.1	1164.40	1164.40	0.00%	0.9	1164.40
RC	15	38	0	0	529.00	529.00	0.00%	2108.8	471.73
RC	15	38	0.5	0.05	978.35	978.35	0.00%	628.4	949.96
RC	15	38	1	0.1	1425.80	1425.80	0.00%	199.7	1425.80
RC	20	51	0	0	692.00	633.00	8.53%	TL	581.74
RC	20	51	0.5	0.05	1252.20	1188.78	5.06%	TL	1152.50
RC	20	51	1	0.1	-	1762.77	-	TL	1762.77
RC	25	64	0	0	-	711.15	-	TL	693.94
RC	25	64	0.5	0.05	-	1485.14	-	TL	1412.96
RC	25	64	1	0.1	2339.10	2224.67	4.89%	TL	2191.20
RC	30	73	0	0	-	1102.66	-	TL	1102.57
RC	30	73	0.5	0.05	2065.70	2065.70	0.00%	566.4	2065.70
RC	30	73	1	0.1	3004.40	3004.40	0.00%	586.5	3004.40
RC	35	89	0	0	-	1043.08	-	TL	1043.39
RC	35	89	0.5	0.05	-	2169.35	-	TL	2106.02
RC	35	89	1	0.1	-	3220.61	-	TL	3151.20
RC	40	100	0	0	-	949.56	-	TL	950.65
RC	40	100	0.5	0.05	-	2156.70	-	TL	2156.70
RC	40	100	1	0.1	-	3116.01	-	TL	3104.69
RC	5	5	0	0	169.00	169.00	0.00%	0.2	169.00
RC	5	5	0.5	0.05	244.95	244.95	0.00%	0.0	244.95
RC	5	5	1	0.1	320.90	320.90	0.00%	0.1	320.90
RC	10	10	0	0	284.00	284.00	0.00%	0.2	284.00
RC	10	10	0.5	0.05	435.30	435.30	0.00%	0.1	435.30
RC	10	10	1	0.1	584.60	584.60	0.00%	0.3	584.60
RC	15	15	0	0	463.00	463.00	0.00%	3.6	463.00
RC	15	15	0.5	0.05	742.80	742.80	0.00%	0.8	742.80
RC	15	15	1	0.1	1021.60	1021.60	0.00%	3.6	1021.60
RC	20	20	0	0	540.00	540.00	0.00%	4.5	540.00
RC	20	20	0.5	0.05	948.05	948.05	0.00%	1.1	948.05
RC	20	20	1	0.1	1356.10	1356.10	0.00%	4.7	1356.10
RC	25	25	0	0	565.00	565.00	0.00%	2.2	565.00
RC	25	25	0.5	0.05	1028.60	1028.60	0.00%	0.7	1028.60
RC	25	25	1	0.1	1490.40	1490.40	0.00%	1.9	1490.40
RC	30	30	0	0	839.00	839.00	0.00%	18.3	769.00
RC	30	30	0.5	0.05	1489.65	1489.65	0.00%	6.0	1433.88
RC	30	30	1	0.1	2129.50	2129.50	0.00%	51.8	1991.45
RC	35	35	0	0	934.00	934.00	0.00%	12.2	934.00
RC	35	35	0.5	0.05	1707.95	1707.95	0.00%	3.9	1707.95
RC	35	35	1	0.1	2475.80	2475.80	0.00%	12.4	2475.80
RC	40	40	0	0	931.00	931.00	0.00%	42.1	931.00
RC	40	40	0.5	0.05	1642.80	1642.80	0.00%	12.5	1642.80
RC	40	40	1	0.1	2352.60	2352.60	0.00%	81.7	2352.60
RC	45	45	0	0	1102.00	1102.00	0.00%	1601.9	1076.11
RC	45	45	0.5	0.05	1984.75	1984.75	0.00%	55.0	1984.75
RC	45	45	1	0.1	2860.40	2860.40	0.00%	205.3	2860.40
RC	50	50	0	0	1062.00	1062.00	0.00%	388.6	1053.18
RC	50	50	0.5	0.05	2010.85	2010.85	0.00%	60.1	1989.19
RC	50	50	1	0.1	2948.10	2948.10	0.00%	195.2	2926.70

Table A.10
Detailed computational results of the temperature-flow formulation of the MC-VRP-TA on the C instances.

Network	$\lvert\mathcal{V}'\rvert$	$\lvert\mathcal{B}\rvert$	γ	δ	BUB	BLB	Gap	Time (s)	RN-LB
C	5	11	0.5	0.05	224.95	224.95	0.00%	0.0	224.95
C	5	11	1	0.1	320.90	320.90	0.00%	0.0	320.90
C	10	24	0.5	0.05	377.15	377.15	0.00%	12.4	366.95
C	10	24	1	0.1	568.40	568.40	0.00%	27.4	552.80
C	15	39	0.5	0.05	692.00	691.90	0.01%	390.8	668.45
C	15	39	1	0.1	1031.60	1031.40	0.01%	1196.1	990.50
C	20	51	0.5	0.05	868.10	843.83	2.79%	TL	821.30
C	20	51	1	0.1	1301.60	1276.81	1.90%	TL	1238.10
C	25	61	0.5	0.05	945.05	880.90	6.78%	TL	868.65
C	25	61	1	0.1	1462.10	1350.49	7.63%	TL	1317.90
C	30	75	0.5	0.05	1124.30	1024.56	8.87%	TL	1009.15
C	30	75	1	0.1	1730.80	1604.15	7.31%	TL	1564.10
C	35	88	0.5	0.05	1565.60	1428.02	8.79%	TL	1377.70
C	35	88	1	0.1	2462.00	2203.87	10.48%	TL	2090.40
C	40	96	0.5	0.05	-	1507.95	-	TL	1439.80
C	40	96	1	0.1	2945.40	2344.09	20.41%	TL	2199.40
C	5	5	0.5	0.05	180.85	180.85	0.00%	0.0	180.85
C	5	5	1	0.1	245.70	245.70	0.00%	0.0	245.70
C	10	10	0.5	0.05	243.70	243.70	0.00%	0.1	243.70
C	10	10	1	0.1	358.40	358.34	0.00%	0.2	358.34
C	15	15	0.5	0.05	460.15	460.15	0.00%	0.0	460.15
C	15	15	1	0.1	637.30	637.30	0.00%	0.0	637.30
C	20	20	0.5	0.05	601.80	601.80	0.00%	0.3	601.80
C	20	20	1	0.1	868.60	868.60	0.00%	0.1	868.60
C	25	25	0.5	0.05	612.25	612.25	0.00%	248.5	576.00
C	25	25	1	0.1	907.60	907.60	0.00%	303.5	865.90
C	30	30	0.5	0.05	734.85	734.85	0.00%	30.3	720.35
C	30	30	1	0.1	1107.20	1107.20	0.00%	34.2	1079.60
C	35	35	0.5	0.05	861.40	861.31	0.01%	1.6	861.31
C	35	35	1	0.1	1281.80	1281.80	0.00%	15.8	1251.10
C	40	40	0.5	0.05	986.75	986.75	0.00%	2.1	984.10
C	40	40	1	0.1	1477.20	1477.20	0.00%	3.2	1467.30
C	45	45	0.5	0.05	1178.80	1163.06	1.33%	TL	1134.65
C	45	45	1	0.1	1780.60	1752.55	1.57%	TL	1719.10
C	50	50	0.5	0.05	1257.80	1221.25	2.91%	TL	1187.15
C	50	50	1	0.1	1894.60	1849.94	2.35%	TL	1780.20

Table A.11
Detailed computational results of the temperature-flow formulation of the MC-VRP-TA on the R instances.

Network	$\lvert\mathcal{V}'\rvert$	$\lvert\mathcal{B}\rvert$	γ	δ	BUB	BLB	Gap	Time (s)	RN-LB
R	5	14	0.5	0.05	567.50	567.50	0.00%	0.2	567.50
R	5	14	1	0.1	823.00	823.00	0.00%	0.2	823.00
R	10	25	0.5	0.05	822.25	822.25	0.00%	19.2	805.05
R	10	25	1	0.1	1166.50	1166.34	0.01%	32.8	1137.90
R	15	35	0.5	0.05	1149.20	1149.08	0.01%	1.9	1146.75
R	15	35	1	0.1	1662.80	1662.80	0.00%	4.0	1648.30
R	20	43	0.5	0.05	1321.10	1321.10	0.00%	750.4	1276.95
R	20	43	1	0.1	1954.20	1954.20	0.00%	471.4	1900.40
R	25	57	0.5	0.05	1556.90	1556.90	0.00%	95.6	1536.00
R	25	57	1	0.1	2322.10	2322.10	0.00%	208.4	2268.40
R	30	69	0.5	0.05	1711.40	1705.99	0.32%	TL	1633.30
R	30	69	1	0.1	2553.60	2516.24	1.46%	TL	2427.90
R	35	85	0.5	0.05	-	2069.25	-	TL	2030.10
R	35	85	1	0.1	3823.90	3119.40	18.42%	TL	3045.20
R	40	91	0.5	0.05	2574.50	2213.68	14.01%	TL	2183.60
R	40	91	1	0.1	-	3301.42	-	TL	3235.30
R	5	5	0.5	0.05	245.90	245.90	0.00%	0.0	245.90
R	5	5	1	0.1	330.80	330.80	0.00%	0.0	330.80
R	10	10	0.5	0.05	526.05	526.05	0.00%	0.0	526.05
R	10	10	1	0.1	758.10	758.10	0.00%	0.0	758.10
R	15	15	0.5	0.05	715.65	715.65	0.00%	0.1	715.65
R	15	15	1	0.1	1032.30	1032.30	0.00%	0.1	1032.30
R	20	20	0.5	0.05	856.65	856.65	0.00%	0.1	856.65
R	20	20	1	0.1	1221.30	1221.30	0.00%	0.4	1221.30
R	25	25	0.5	0.05	1029.45	1029.37	0.00%	80.5	996.05
R	25	25	1	0.1	1468.90	1468.90	0.00%	23.0	1428.30
R	30	30	0.5	0.05	1128.75	1128.75	0.00%	3.7	1118.35
R	30	30	1	0.1	1625.10	1625.10	0.00%	3.3	1613.70
R	35	35	0.5	0.05	1218.30	1218.30	0.00%	6.2	1206.95
R	35	35	1	0.1	1775.60	1775.60	0.00%	3.1	1759.00
R	40	40	0.5	0.05	1387.00	1343.34	3.15%	TL	1317.00
R	40	40	1	0.1	2019.80	1966.47	2.64%	TL	1923.10
R	45	45	0.5	0.05	1567.70	1567.70	0.00%	368.9	1531.25
R	45	45	1	0.1	2296.40	2296.40	0.00%	793.6	2235.80
R	50	50	0.5	0.05	1661.00	1640.48	1.23%	TL	1610.85
R	50	50	1	0.1	2454.10	2454.10	0.00%	2759.5	2376.30

Table A.12
Detailed computational results of the temperature-flow formulation of the MC-VRP-TA on the RC instances.

Network	$\|\mathcal{V}'\|$	$\|\mathcal{B}\|$	γ	δ	BUB	BLB	Gap	Time (s)	RN-LB
RC	5	12	0.5	0.05	484.10	484.10	0.00%	0.0	484.10
RC	5	12	1	0.1	644.20	644.20	0.00%	0.0	644.20
RC	10	24	0.5	0.05	808.25	808.25	0.00%	39.0	796.00
RC	10	24	1	0.1	1143.50	1143.50	0.00%	21.9	1121.00
RC	15	38	0.5	0.05	977.75	977.75	0.00%	6.2	963.20
RC	15	38	1	0.1	1419.90	1419.90	0.00%	30.1	1339.40
RC	20	51	0.5	0.05	1240.25	1218.63	1.74%	TL	1128.00
RC	20	51	1	0.1	1803.40	1758.27	2.50%	TL	1628.60
RC	25	64	0.5	0.05	1536.90	1533.95	0.19%	TL	1470.80
RC	25	64	1	0.1	-	2251.85	-	TL	2160.20
RC	30	73	0.5	0.05	2051.75	2044.92	0.33%	TL	1947.60
RC	30	73	1	0.1	2980.50	2963.02	0.58%	TL	2895.20
RC	35	89	0.5	0.05	-	2215.64	-	TL	2120.05
RC	35	89	1	0.1	-	3285.97	-	TL	3201.60
RC	40	100	0.5	0.05	-	2295.51	-	TL	2157.50
RC	40	100	1	0.1	4023.50	3408.73	15.28%	TL	3232.00
RC	5	5	0.5	0.05	244.95	244.95	0.00%	0.0	244.95
RC	5	5	1	0.1	320.90	320.90	0.00%	0.0	320.90
RC	10	10	0.5	0.05	430.80	430.80	0.00%	0.0	430.80
RC	10	10	1	0.1	575.60	575.60	0.00%	0.0	575.60
RC	15	15	0.5	0.05	736.25	736.19	0.01%	29.5	688.20
RC	15	15	1	0.1	1008.50	1008.31	0.01%	34.6	944.30
RC	20	20	0.5	0.05	924.10	924.01	0.01%	0.2	924.01
RC	20	20	1	0.1	1308.20	1308.20	0.00%	0.5	1297.40
RC	25	25	0.5	0.05	1007.80	1007.75	0.00%	0.5	1007.75
RC	25	25	1	0.1	1449.40	1449.27	0.01%	0.7	1449.27
RC	30	30	0.5	0.05	1470.15	1367.75	6.97%	TL	1337.45
RC	30	30	1	0.1	2092.30	1938.09	7.37%	TL	1895.00
RC	35	35	0.5	0.05	1697.60	1697.40	0.01%	756.5	1595.85
RC	35	35	1	0.1	2456.10	2456.10	0.00%	57.2	2279.80
RC	40	40	0.5	0.05	1632.10	1632.10	0.00%	10.7	1617.80
RC	40	40	1	0.1	2313.20	2312.93	0.01%	26.1	2263.40
RC	45	45	0.5	0.05	1970.75	1970.75	0.00%	173.2	1923.95
RC	45	45	1	0.1	2833.30	2833.01	0.01%	607.8	2683.30
RC	50	50	0.5	0.05	1992.55	1953.32	1.97%	TL	1854.90
RC	50	50	1	0.1	2905.50	2864.57	1.41%	TL	2755.40

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